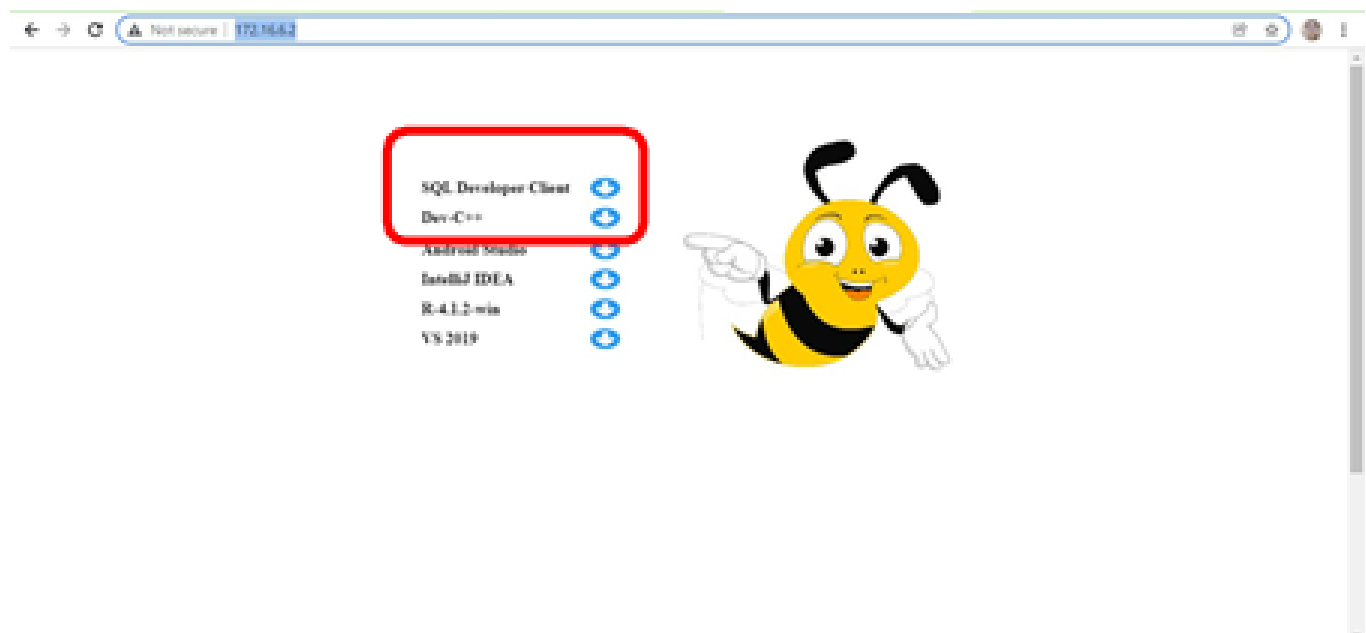


## Objectives

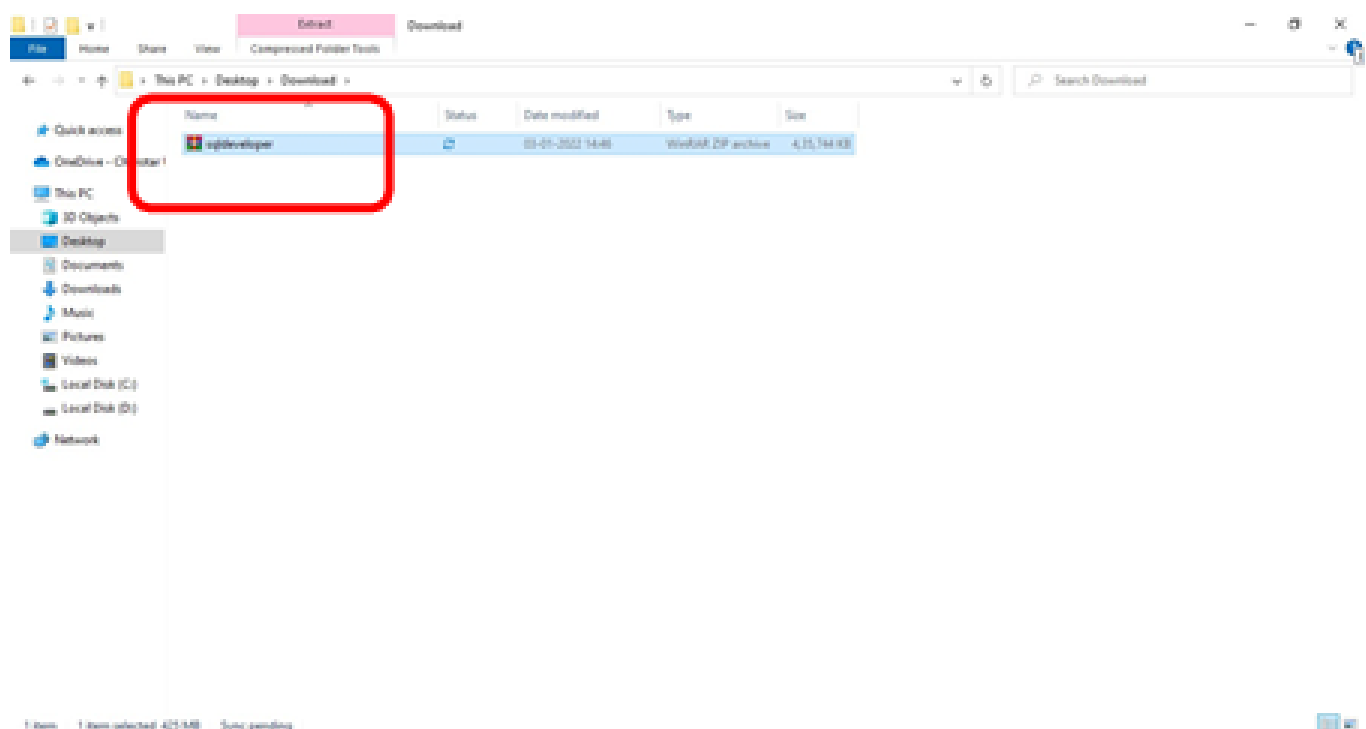
- To make students acquaint with installation of Database Client Software.

**Step: 1** Go to <http://172.16.6.2/>

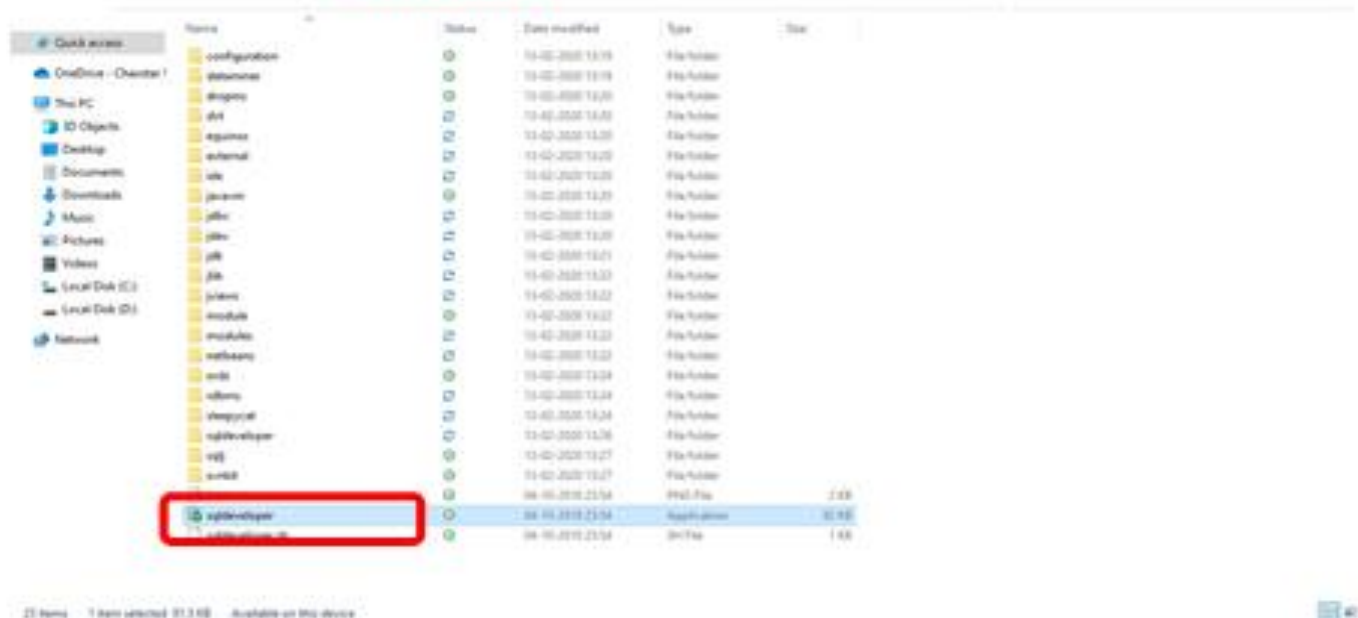
Download SQL Developer Client



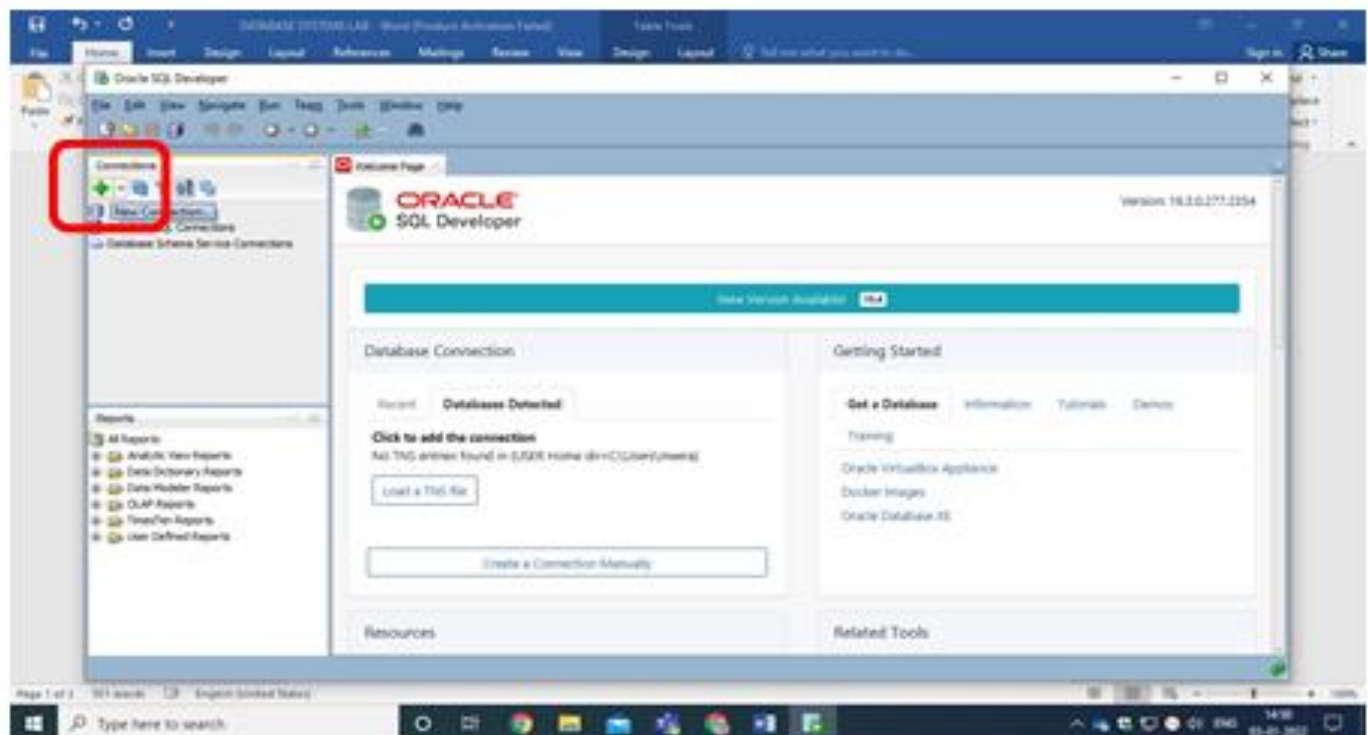
This will download the zip with name sqldeveloper. Unzip it.



**Step: 2** Double click on sqldeveloper Application

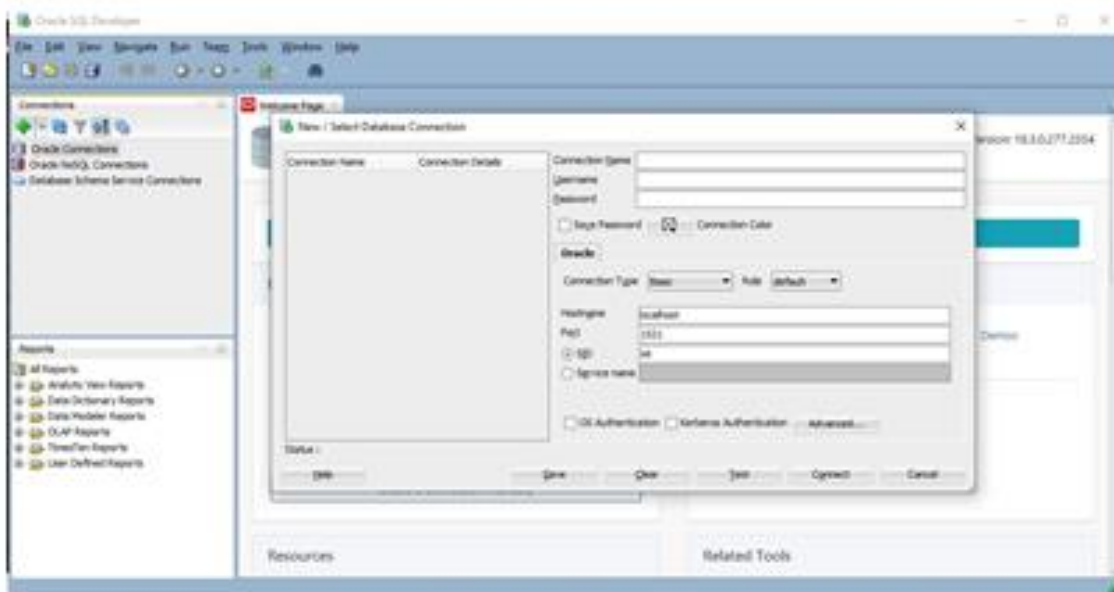


It will open the Oracle SQL Developer.



**Step: 3** Click on  Sign, the following screen will appear.

## Tool Based Manual || BCA || SEM II || CA130 : Database Fundamentals



**Step: 4** Give Proper Connection Name for example CMPICA

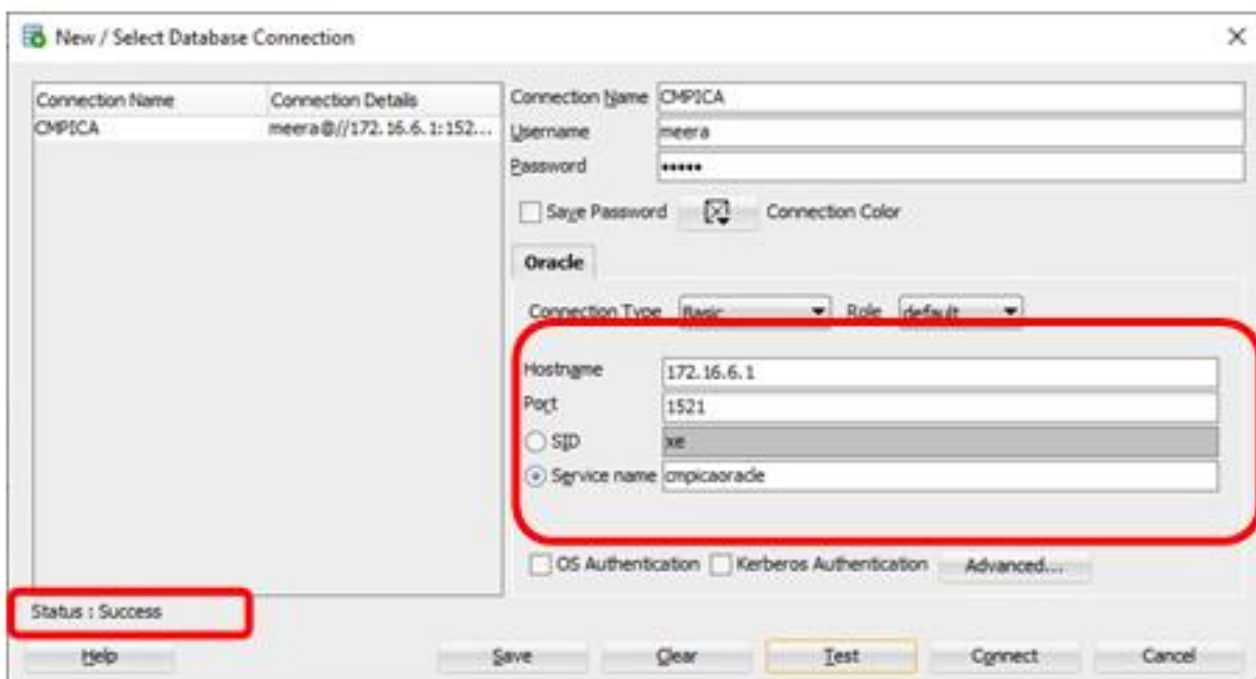
Enter Username and Password provided to you in respective fields.

In Hostname write **172.16.6.1**

Click service name radio button and enter **cmpicaoracle**

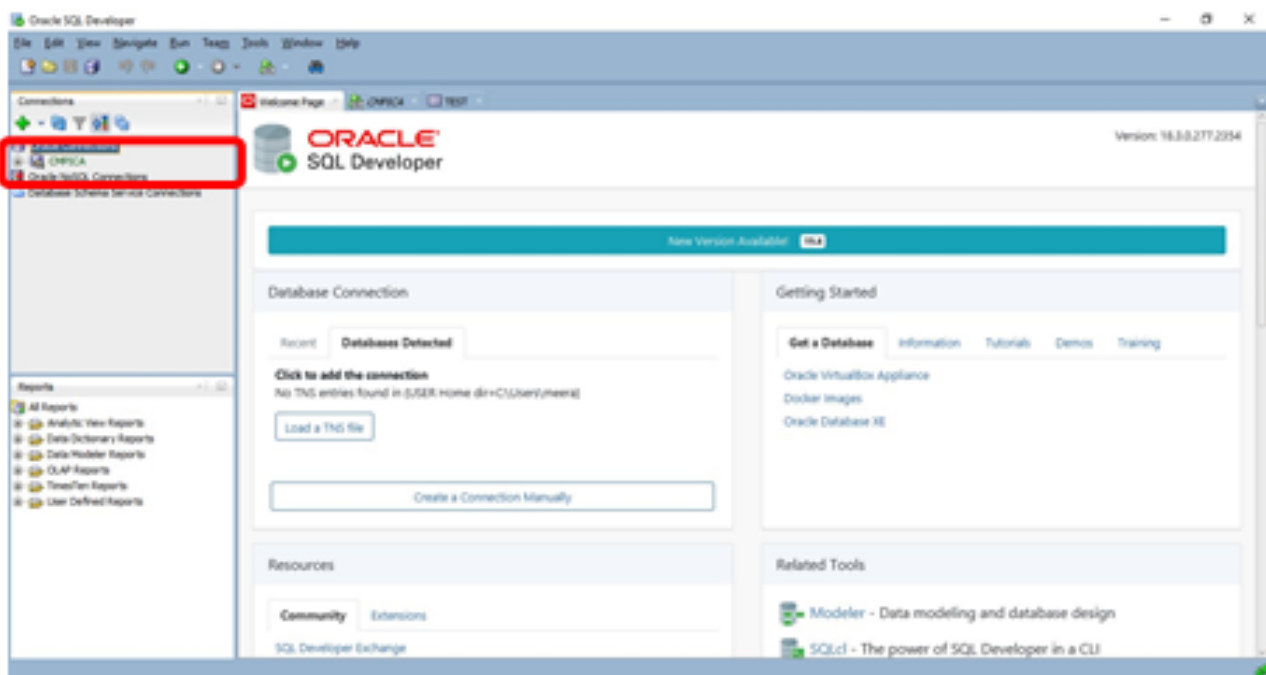
KEEP REST OF THE CONFIGURATION AS IT IS. And Click on Test button. The **SUCCESS** should get displayed.

Now click on Connect Button.

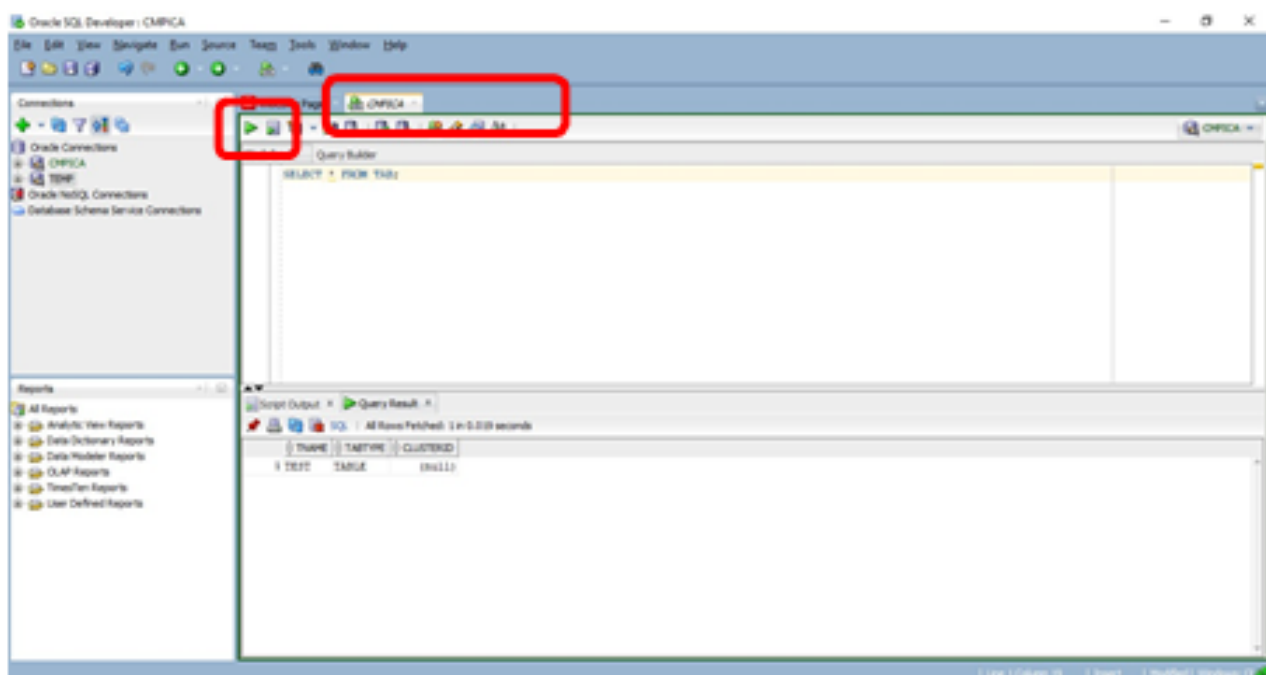


Your newly created connection should get displayed on the left panel.

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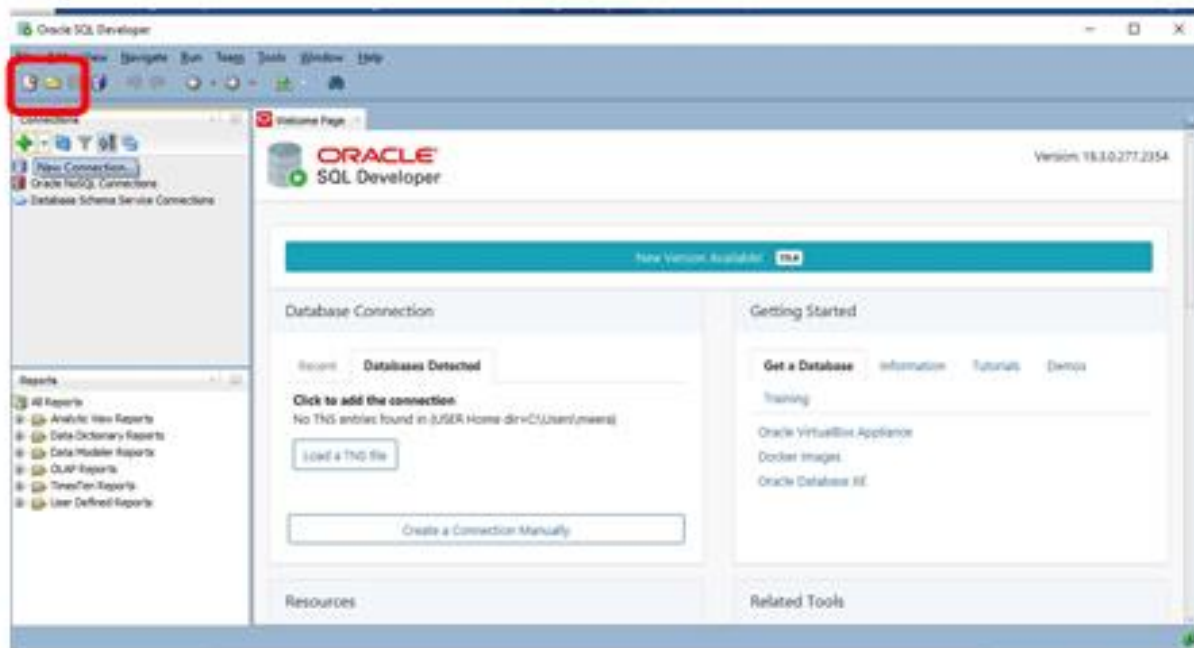
**Step 5** On the right side panel observe the Tab with CMPICA (Connection Name). Write your query and Run your query (execute) using  icon **OR** press Ctrl + Enter.



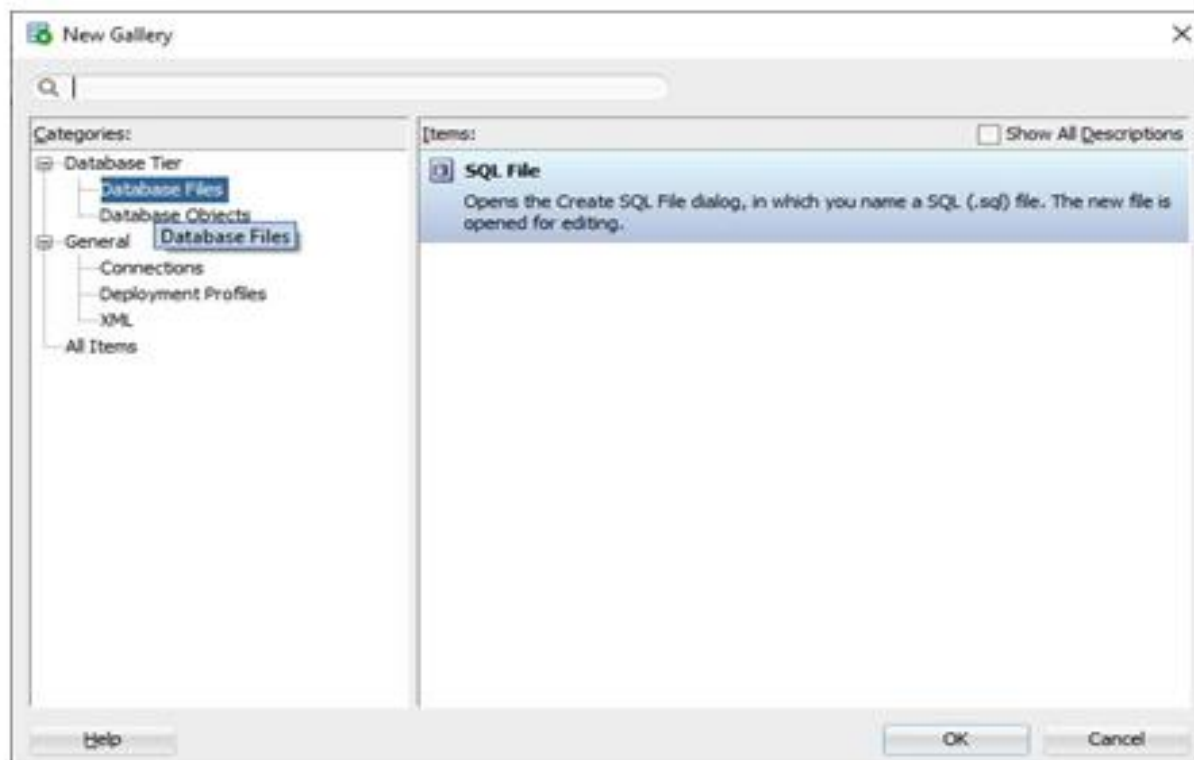
**Step 6** IF you want to create a new SQL file then

In the right panel click on File -> New symbol,  or press Ctrl + N and Then select database -> SQL File and click OK. Please refer below screenshots.

## Tool Based Manual || BCA || SEM II || CA130 : Database Fundamentals



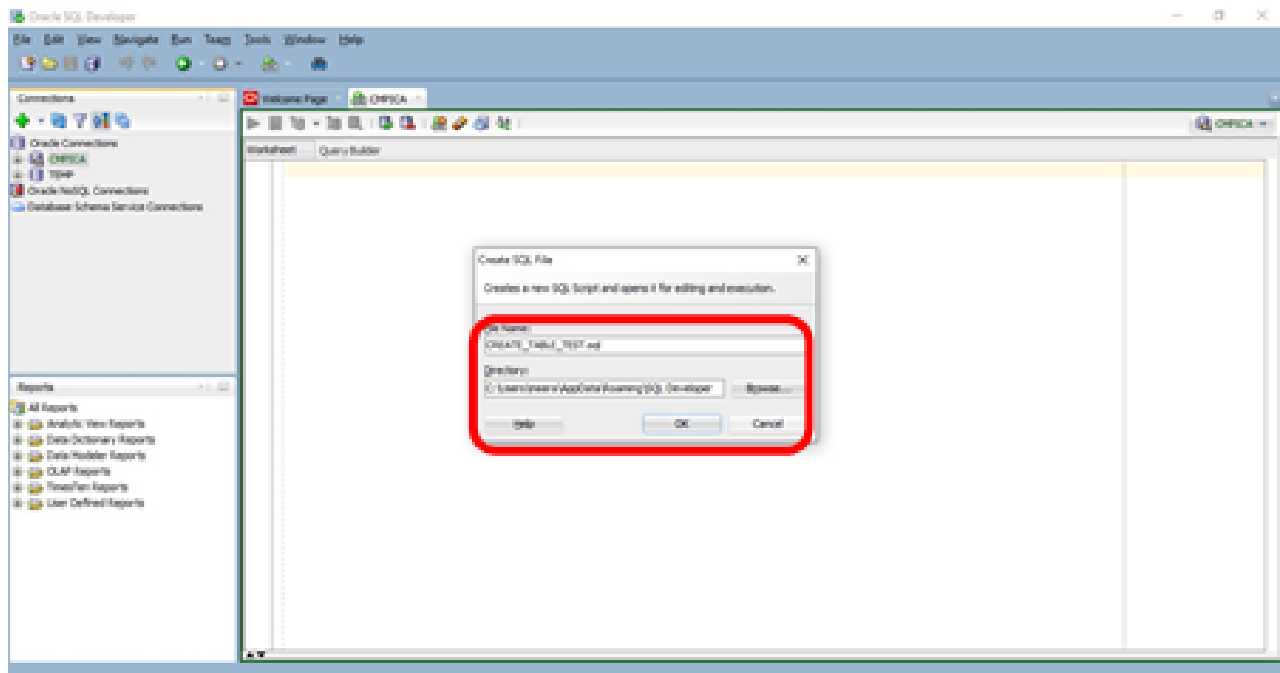
The following screen will appear.



Give proper file name and save to the proper location. [IT IS ADVISED TO SAVE ALL THE SQL IN ONE FOLDER]

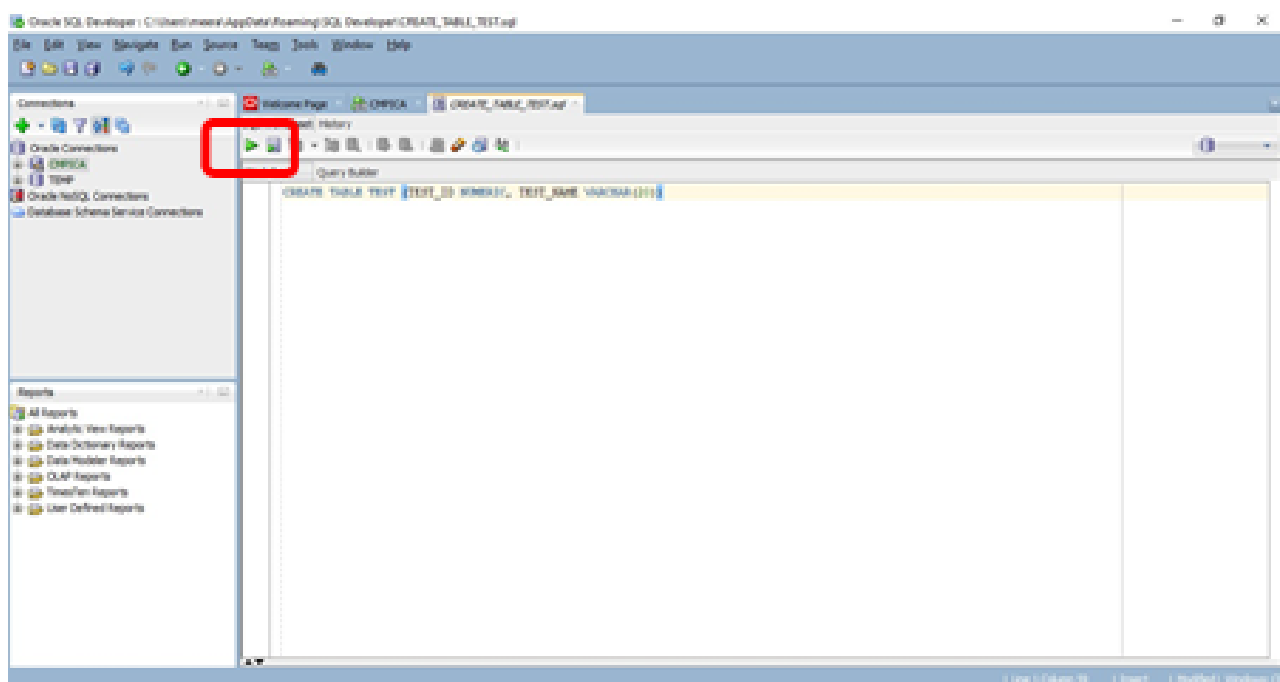


## Tool Based Manual || BCA || SEM II || CA130 : Database Fundamentals



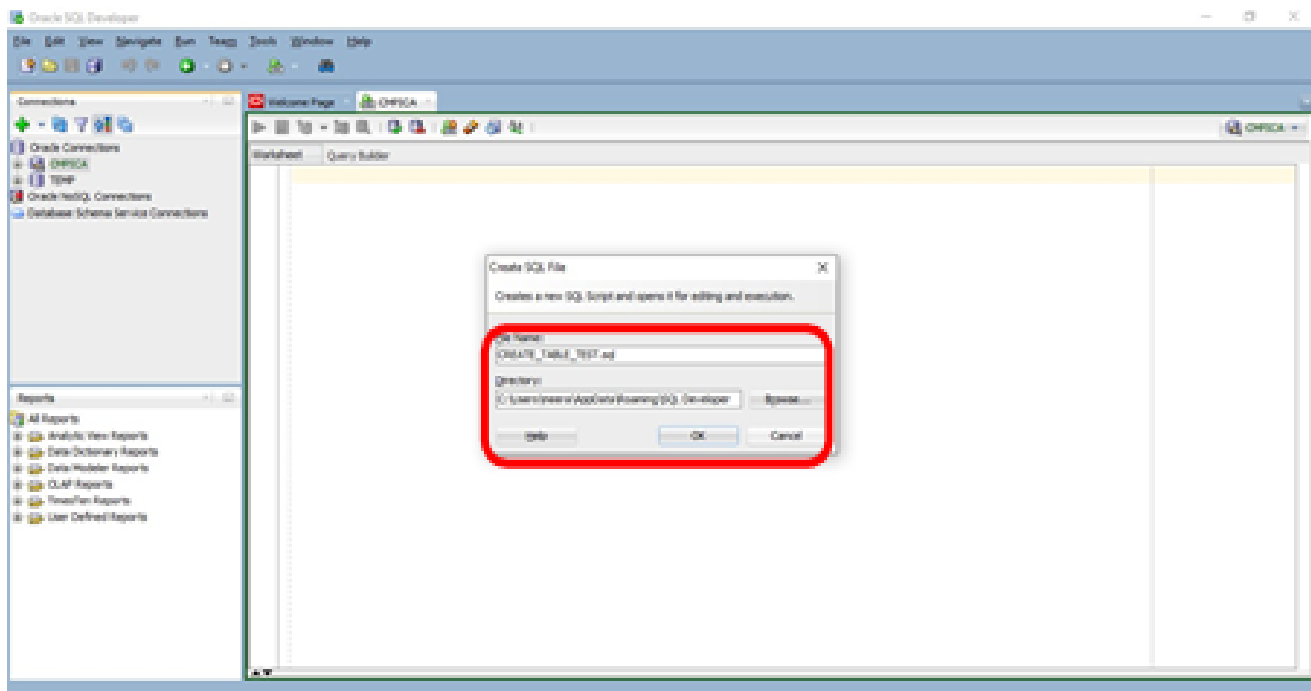
Click OK.

**Step 7** Write your query and Run your query (execute) using  icon **OR** press Ctrl + Enter. Select the appropriate Connection in the dialogue box and press OK.



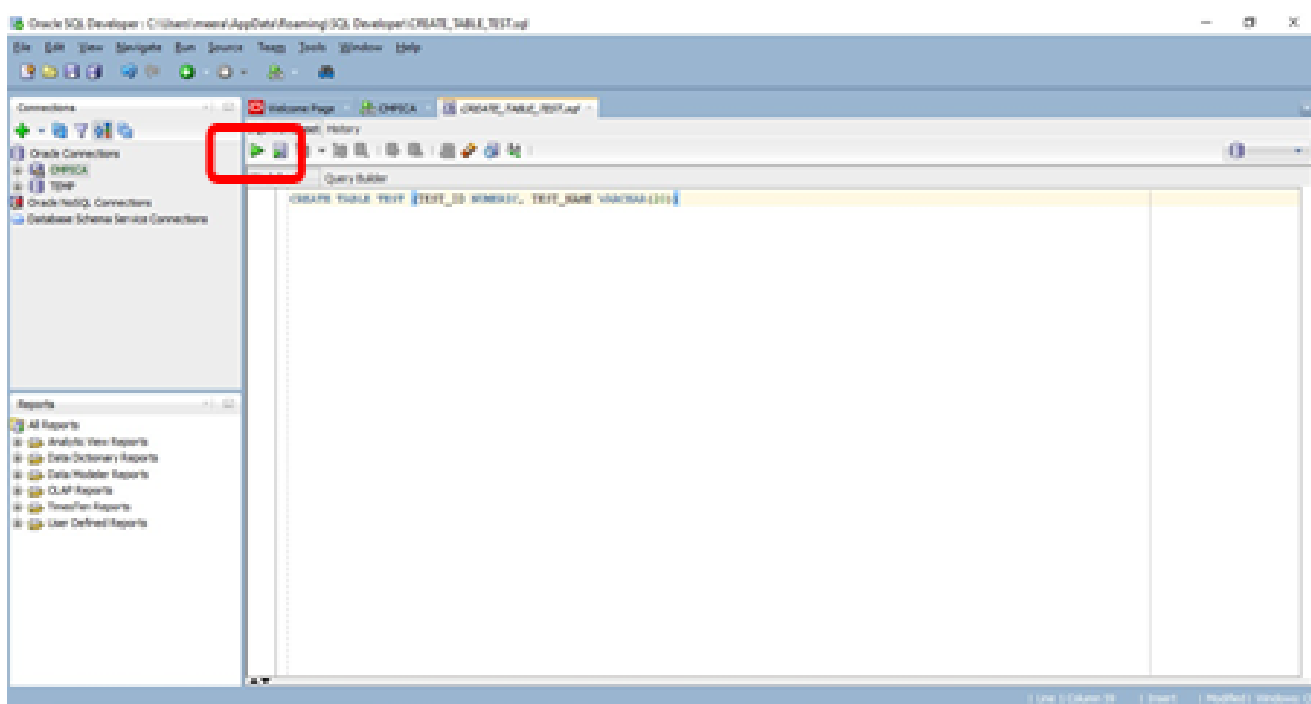
Select the appropriate connection and press OK.

## Tool Based Manual || BCA || SEM II || CA130 : Database Fundamentals



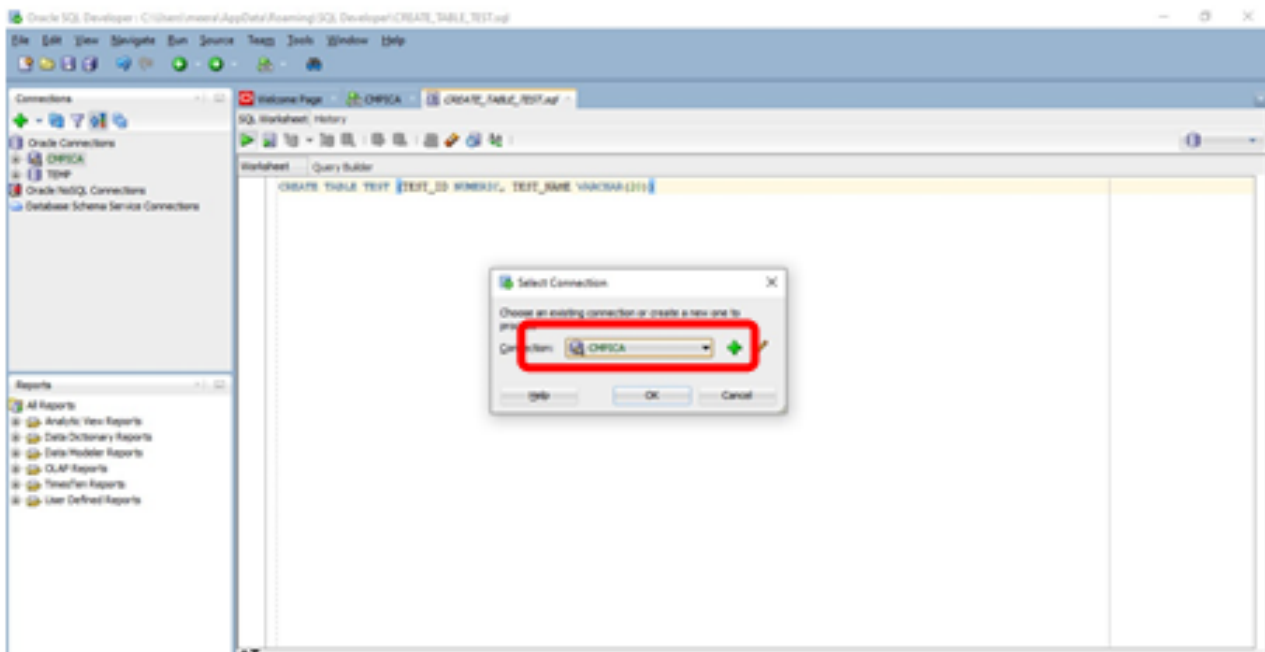
Click OK.

**Step 7** Write your query and Run your query (execute) using  icon **OR** press Ctrl + Enter. Select the appropriate Connection in the dialogue box and press OK.

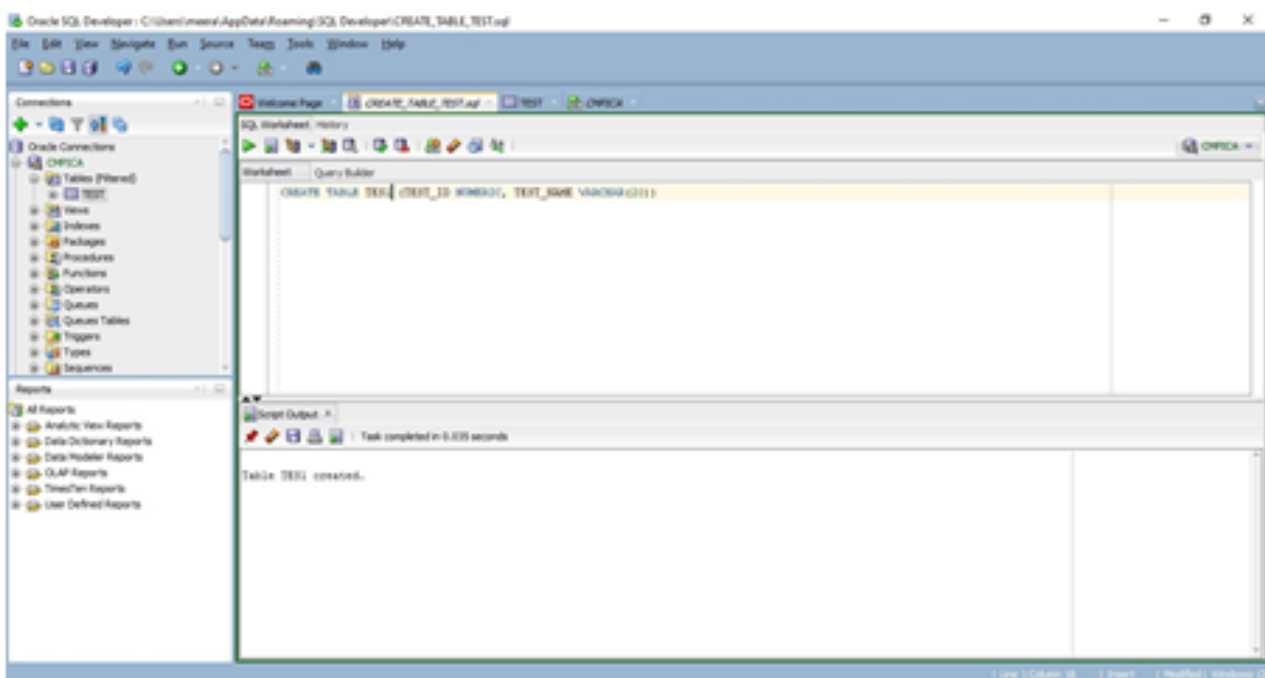


Select the appropriate connection and press OK.

## Tool Based Manual || BCA || SEM II || CA130 : Database Fundamentals



Observe execution result in bottom panel.



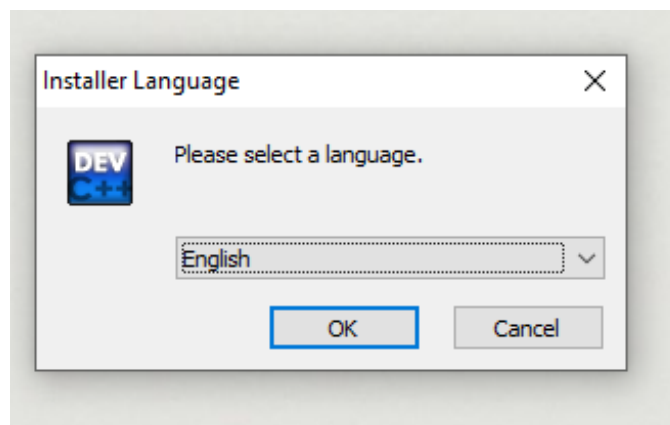
**Congratulations!!! Your Oracle SQL Developer Setup is completed successfully.**



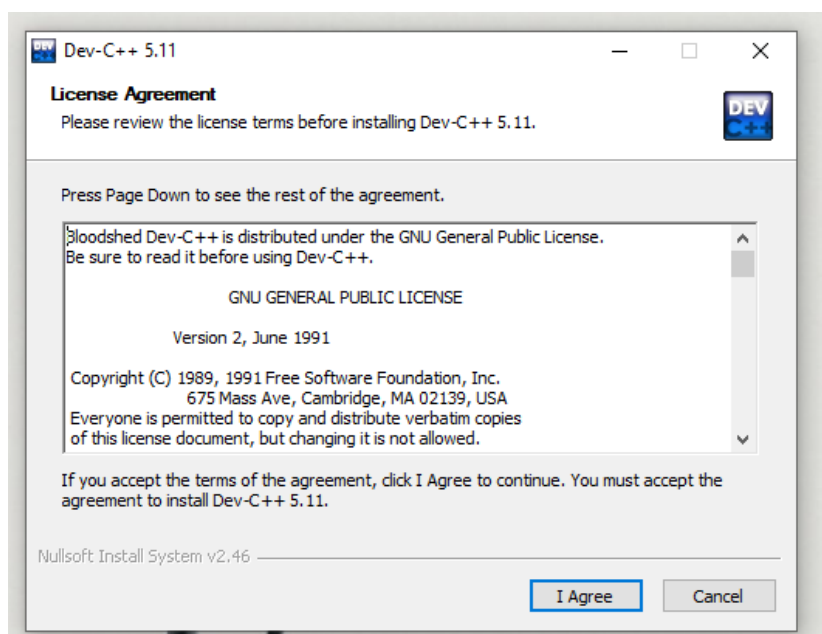
## Installation and Getting Started under Windows OS

Follow the steps mentioned in following figure to install Dev c++ on Windows Operating System.

1. Go to <https://sourceforge.net/projects/orwelldvcpp/>
2. Click on Download and wait for five seconds.
3. Double click on Dev-Cpp 5.11 TDM-GCC 4.9.2 Setup file

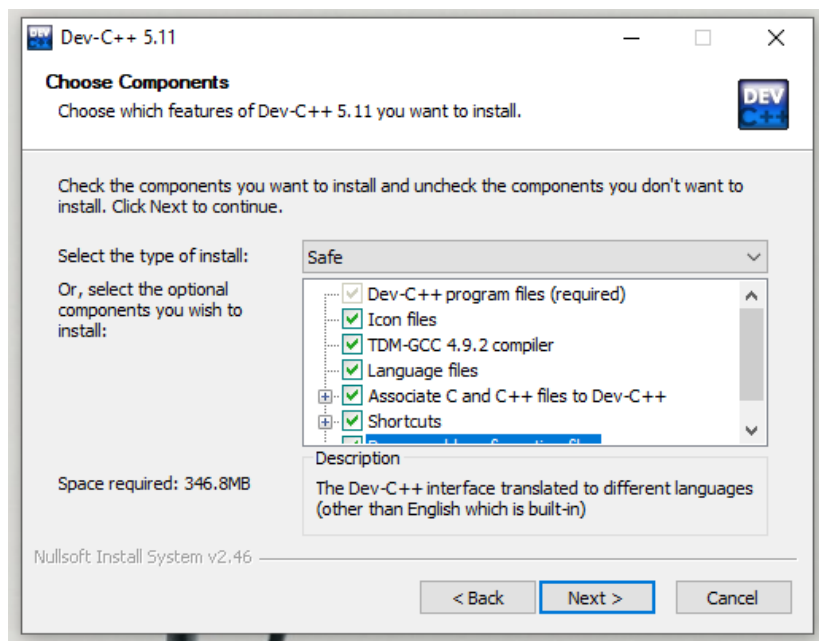


4. Click On Ok.

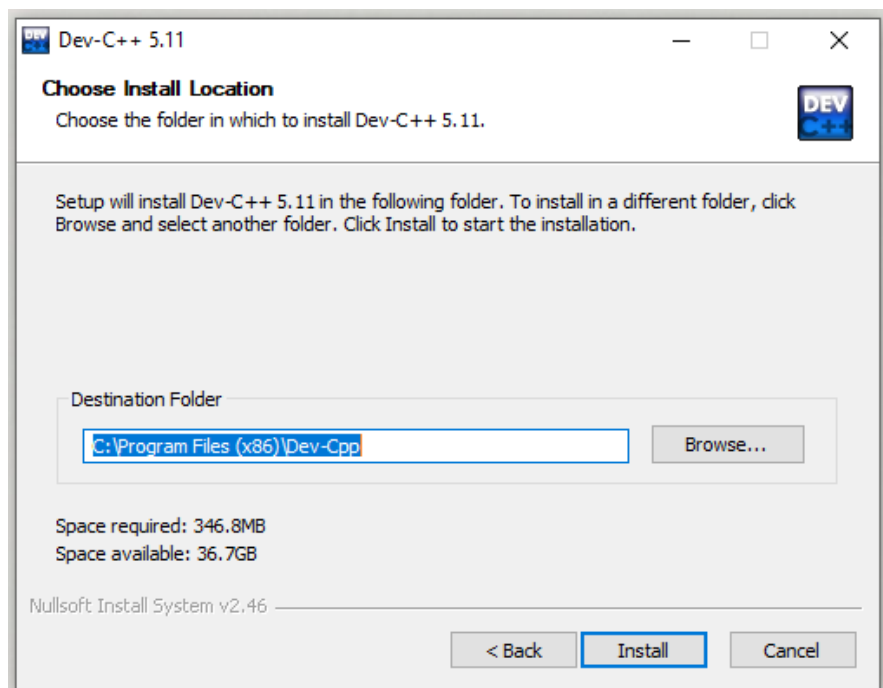


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5. Click On I Agree.

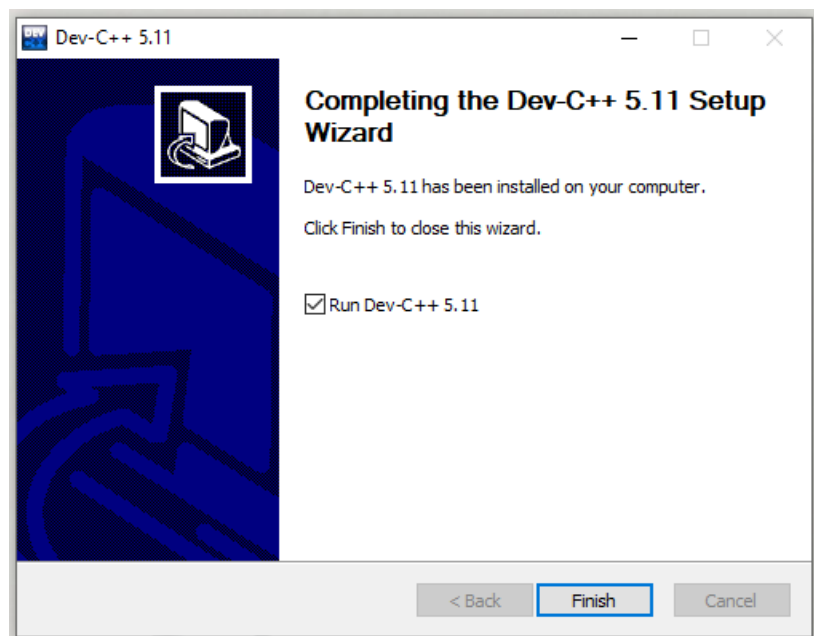


6. Click on Next.

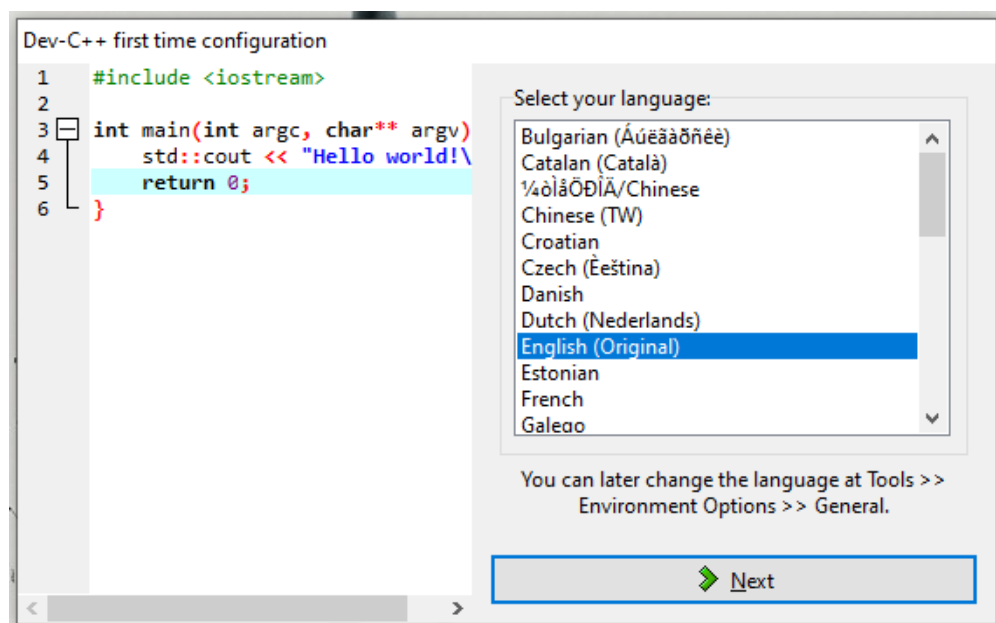


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7. Click on Install.

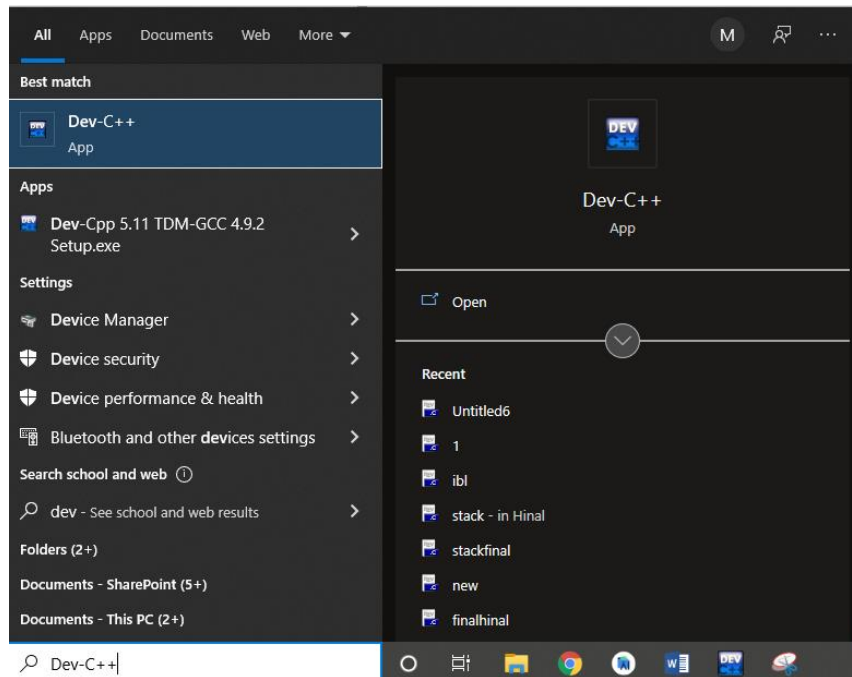


8. Click on finish.

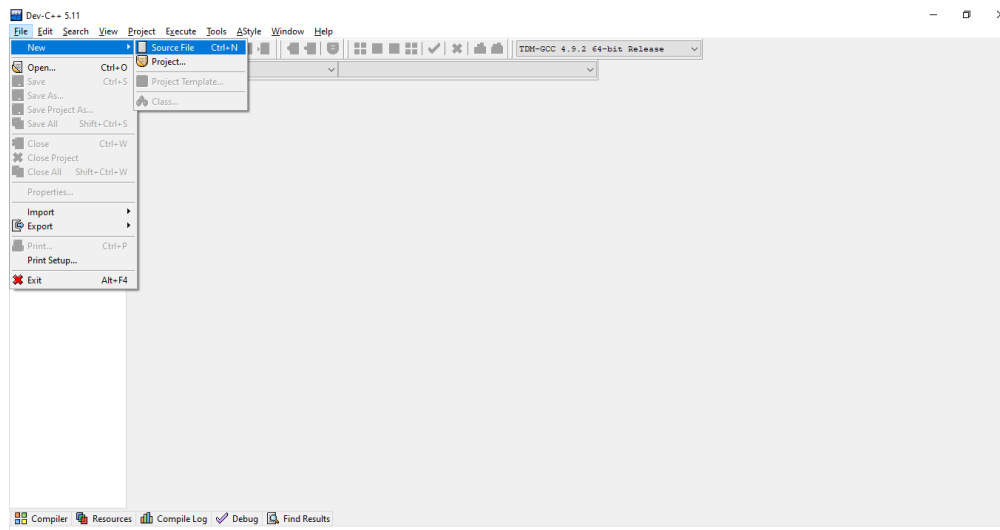


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9. Click on Next, Next Okay.

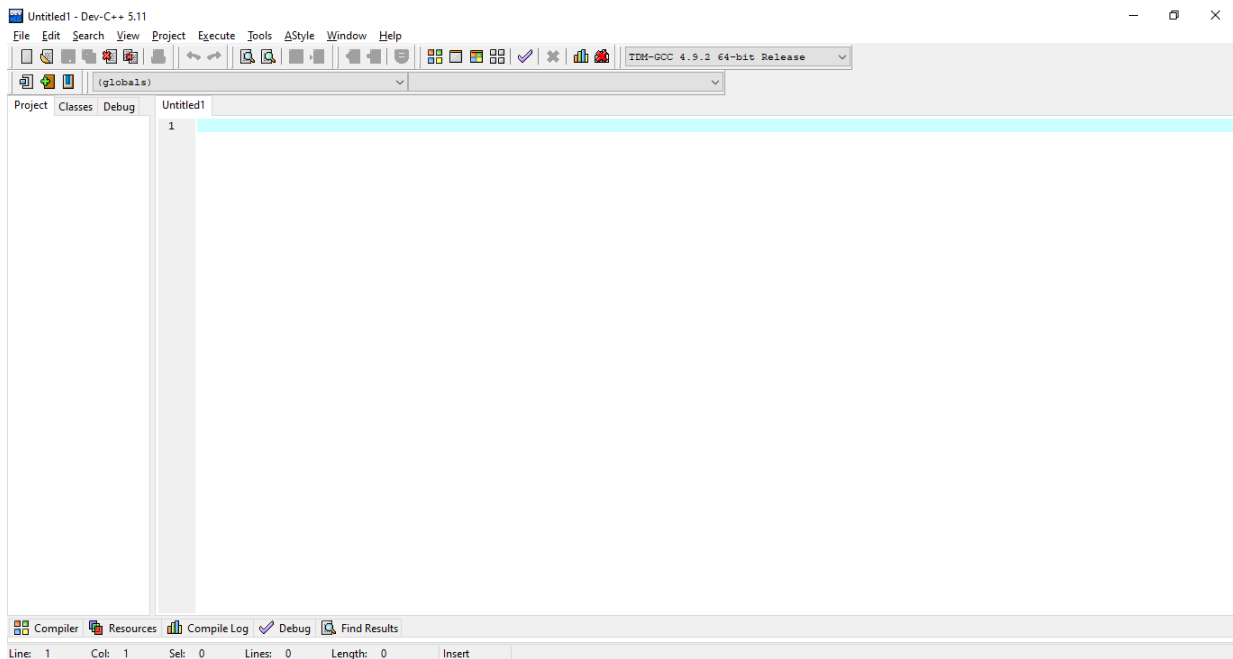


10. Open Dev C++.

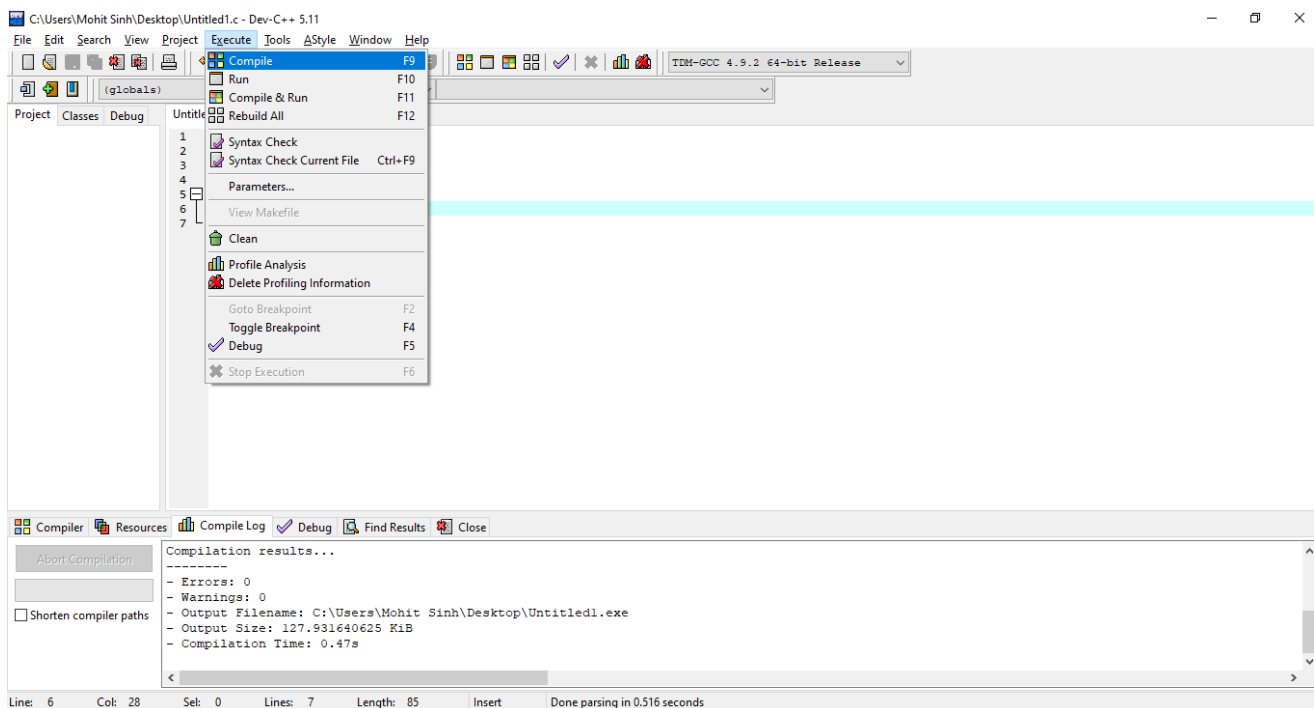


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### 11. IDE of Dev C++.

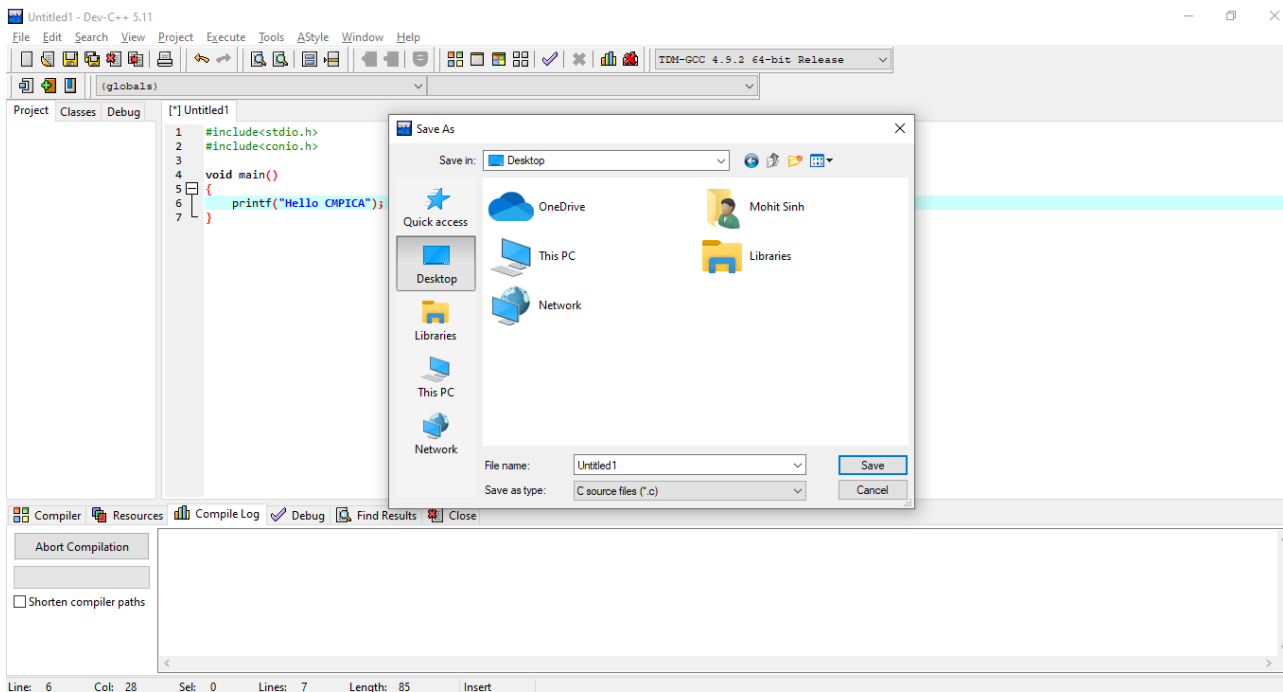


### 12. New file for c programming.

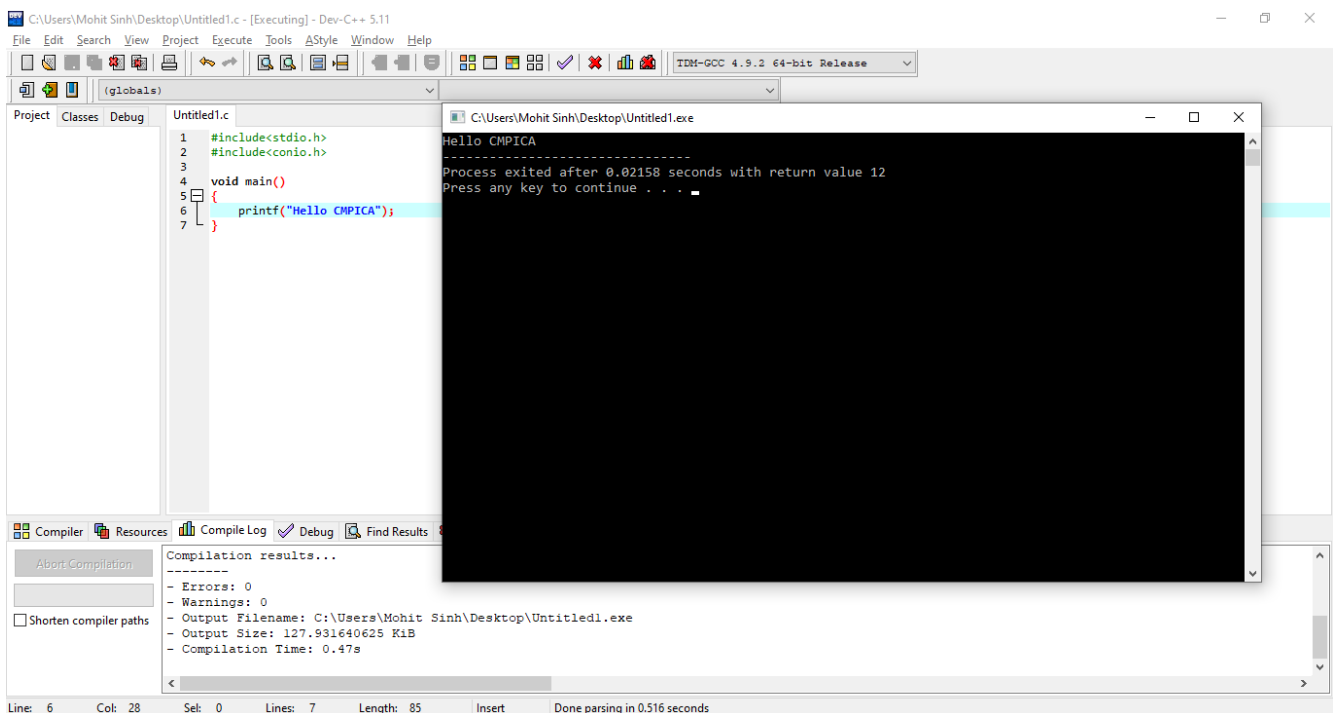


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13. From the execute tab of menu bar you have to select compile option.



14. To run the program, save the file with appropriate name and extension.



The black screen window is the console window that display the output of the program.





## Setup visual studio 2022 and Web application with ASP.NET MVC 6.0

.NET  
Core

In this tutorial, I'll show you how to download and install Visual Studio 2022 and construct a simple.NET Console application step by step.



Version 17.0 of Visual Studio 2022 supports applications written with .Net 6.0. We have downloaded and installed Visual Studio 2022 Community Edition Version 17.0. The Community edition of Visual Studio 2022 is available for free.

So, let's start

## Step 1

You can download Visual Studio 2022 version 17.0 from this [link](#).

## Downloads

The screenshot shows the Visual Studio 2022 download page. On the left, there is a section for 'Visual Studio 2022' with the version '17.0' and a description: 'The best comprehensive IDE for .NET and C++ developers on Windows. Fully packed with a sweet array of tools and features to elevate and enhance every stage of software development.' Below this are links for 'Release notes', 'Compare Editions', and 'How to install offline'. To the right, there are four columns for different editions: 'Community' (Powerful IDE, free for students, open-source contributors, and individuals, with a 'Free download' button), 'Professional' (Professional IDE best suited to small teams, with a 'Free trial' button), 'Enterprise' (Scalable, end-to-end solution for teams of any size, with a 'Free trial' button), and 'Preview' (Get early access to latest features not yet in the main release, with 'Learn more' and 'Release notes' links).

## Step 2

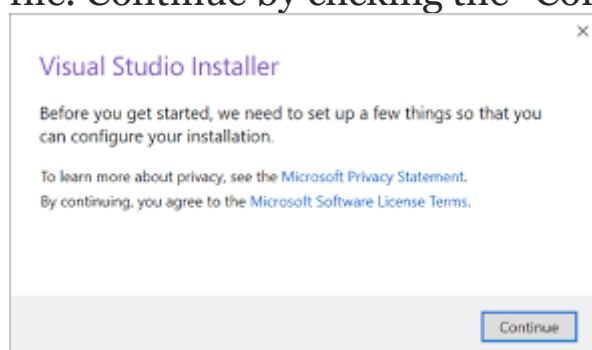
To download the VS 2022 executable file to the downloads directory, click the “Download” button.



The screenshot shows the Microsoft Visual Studio download page. At the top, there's a navigation bar with links for Microsoft, Visual Studio, Products, Downloads, Buy, Support, and Subscriber Access. A 'Free Visual Studio' button is also present. Below the navigation bar, a large purple banner reads 'Thank you for downloading Visual Studio' and 'Your download will start shortly. If your download does not begin, [click here to retry](#)'. On the left, there's a 'Getting started' dropdown menu and a 'Follow us' section with social media icons and a link to sign up for the developer community newsletter. The main content area has a section titled 'Install Visual Studio' with a sub-header 'Getting Started with Visual Studio IDE'. Below this, there's a video player showing 'Starting with Visual Studio' and a text block explaining that the Visual Studio IDE brings together graphical designers, compilers, code completion tools, source control, extensions, and many more features in one place.

## Step 3

The Visual Studio installer window will open when you double-click the.exe file. Continue by clicking the “Continue” button.



## Step 4

After hitting the Continue button, the download and installation progress bar window will appear.

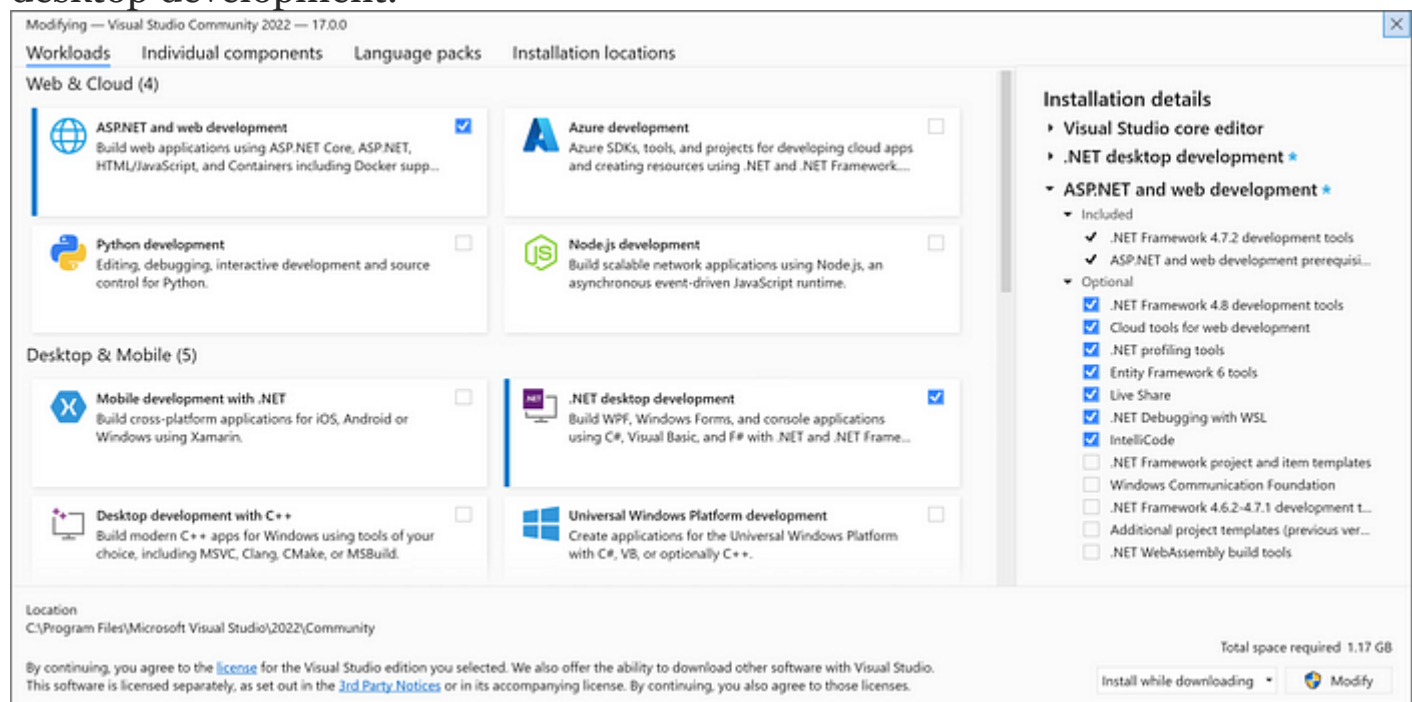
## Visual Studio Installer

Almost done ... Getting everything ready.



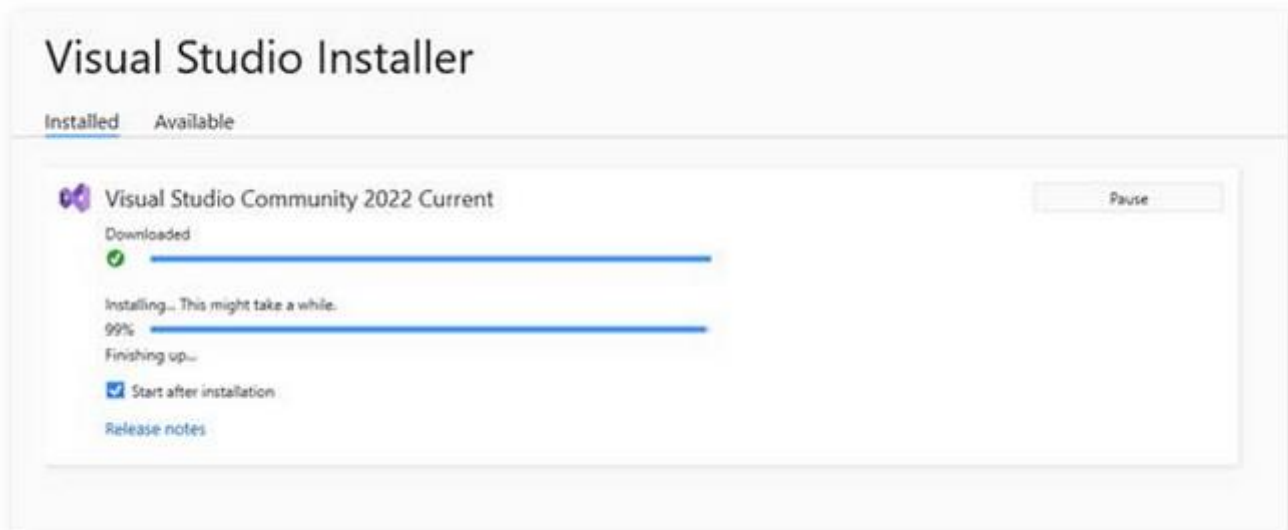
## Step 5

Workloads will open after the download and installation are completed. We must decide which workloads we require. We chose ASP.NET and .NET desktop development.



## Step 6

After selecting the appropriate packages, simply click the “Install” button.



When the installation is finished and we open Visual Studio for the first time, it takes 15–30 seconds for its initial internal setup.



We're getting things ready

Thanks for your patience.

.....

Your existing Visual Studio profile will be loaded. It may also request that you sign in to your Microsoft account. You can establish a new Microsoft account if you don't already have one.





The sign-in process is optional, so you can skip it. But don't forget to do this! Log up with your Gmail or Microsoft account to activate Visual Studio and receive a free lifetime licence.

If you don't have an account yet, select Not now, perhaps later.

## Visual Studio

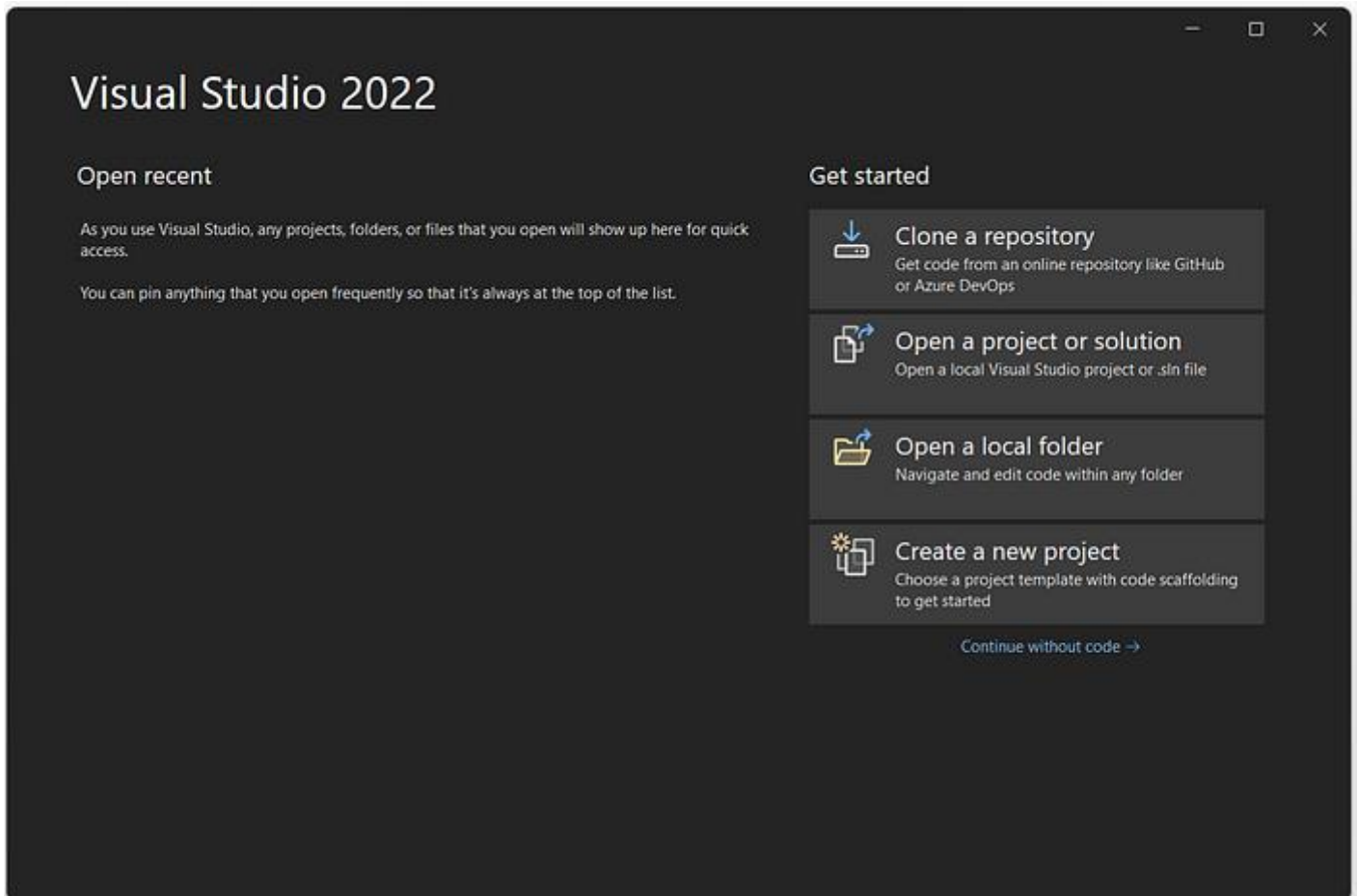
### Sign in to Visual Studio

- Sync settings across devices
- Collaborate in real time with LiveShare
- Integrate seamlessly with Azure Services

Sign in

No account? [Create one!](#)

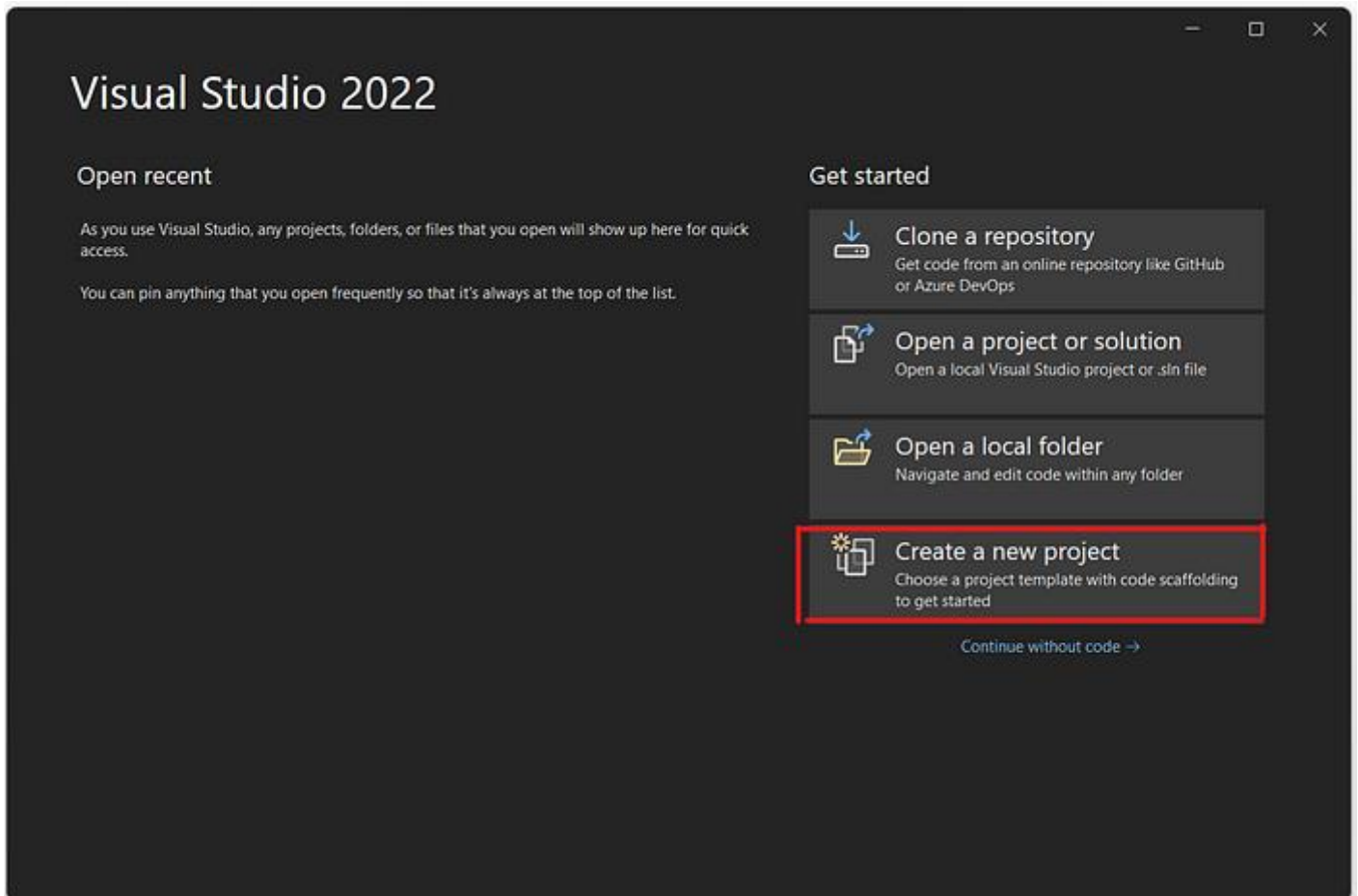
The installation of Visual Studio 2022 is now complete, and you can use it. We can observe a newly revamped Visual Studio landing page with a list of recent projects. We can also use GitHub and Azure DevOps to clone or check out the code. A project or solution, as well as a folder, can be opened. We can also start a new project.



## Creating Your First Project

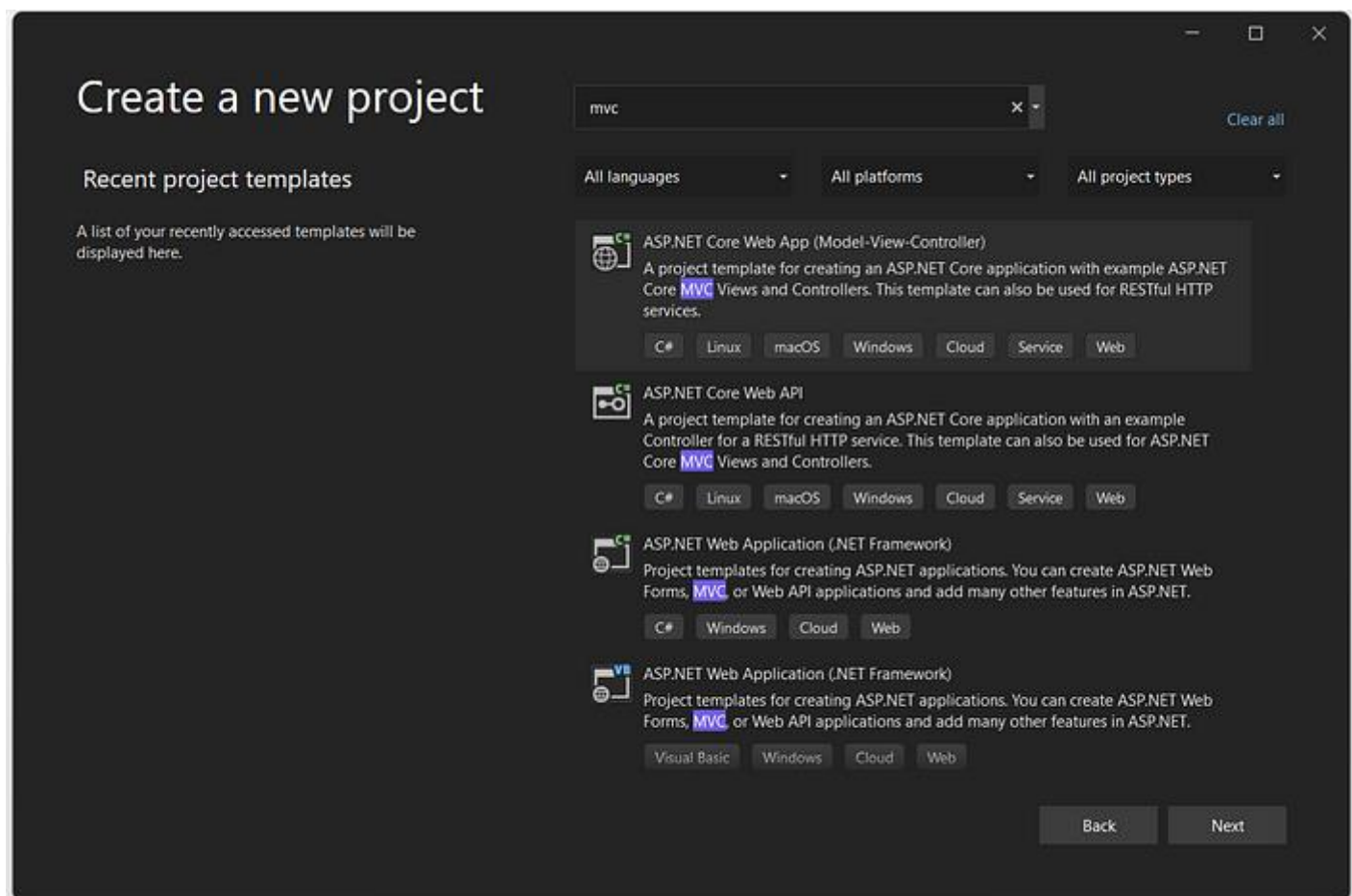
### Step1

Click on Create a new Project to create a new project.

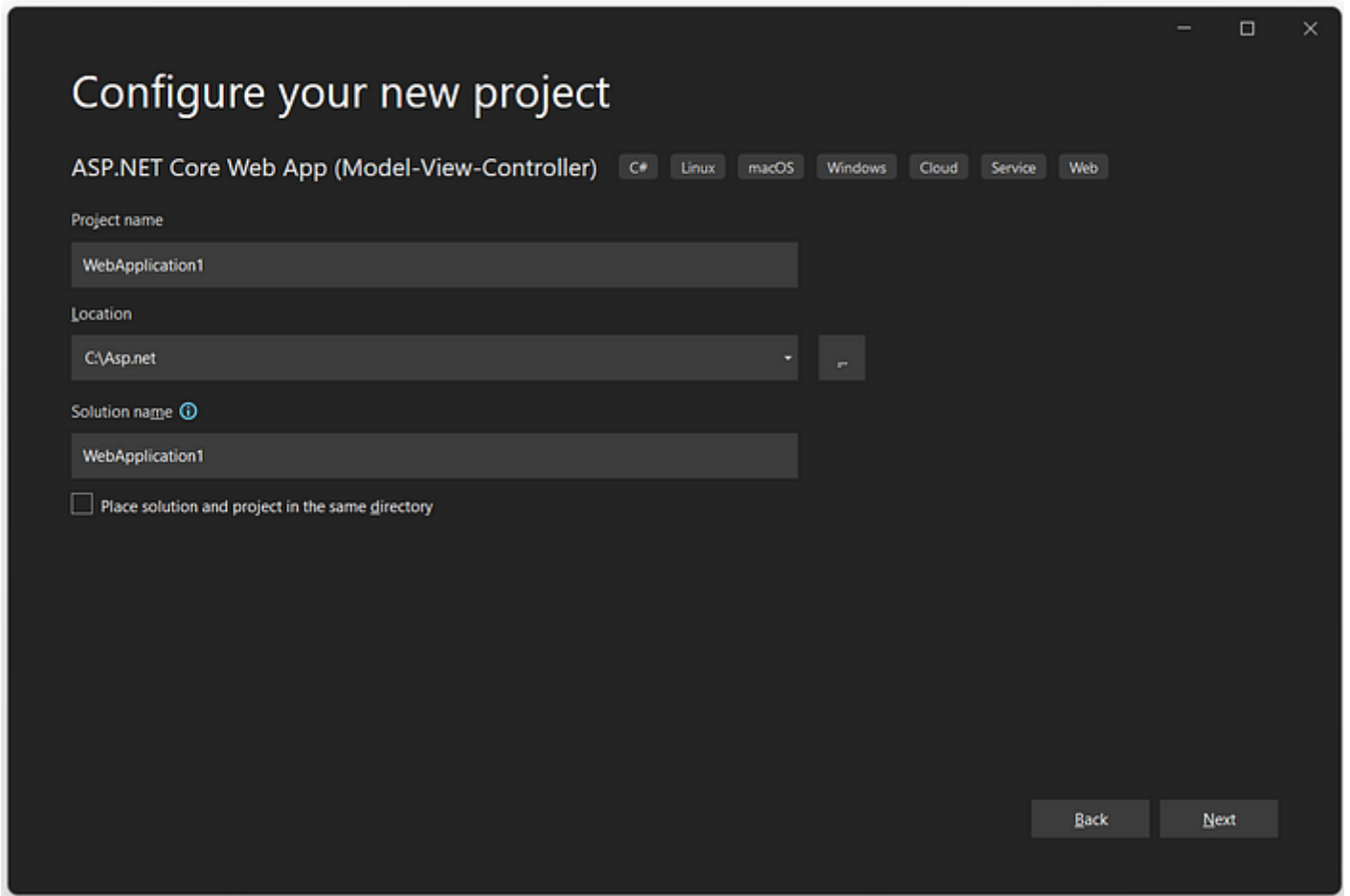


There are a variety of project types to choose from. For the time being, select the “Asp.Net Core MVC Web Application” project type.

Click Next after selecting Asp.Net Core MVC Application.



Give your project a meaningful name and select a path for it. Then click Next button.



Configure your new project

ASP.NET Core Web App (Model-View-Controller) C# Linux macOS Windows Cloud Service Web

Project name

WebApplication1

Location

C:\Asp.net

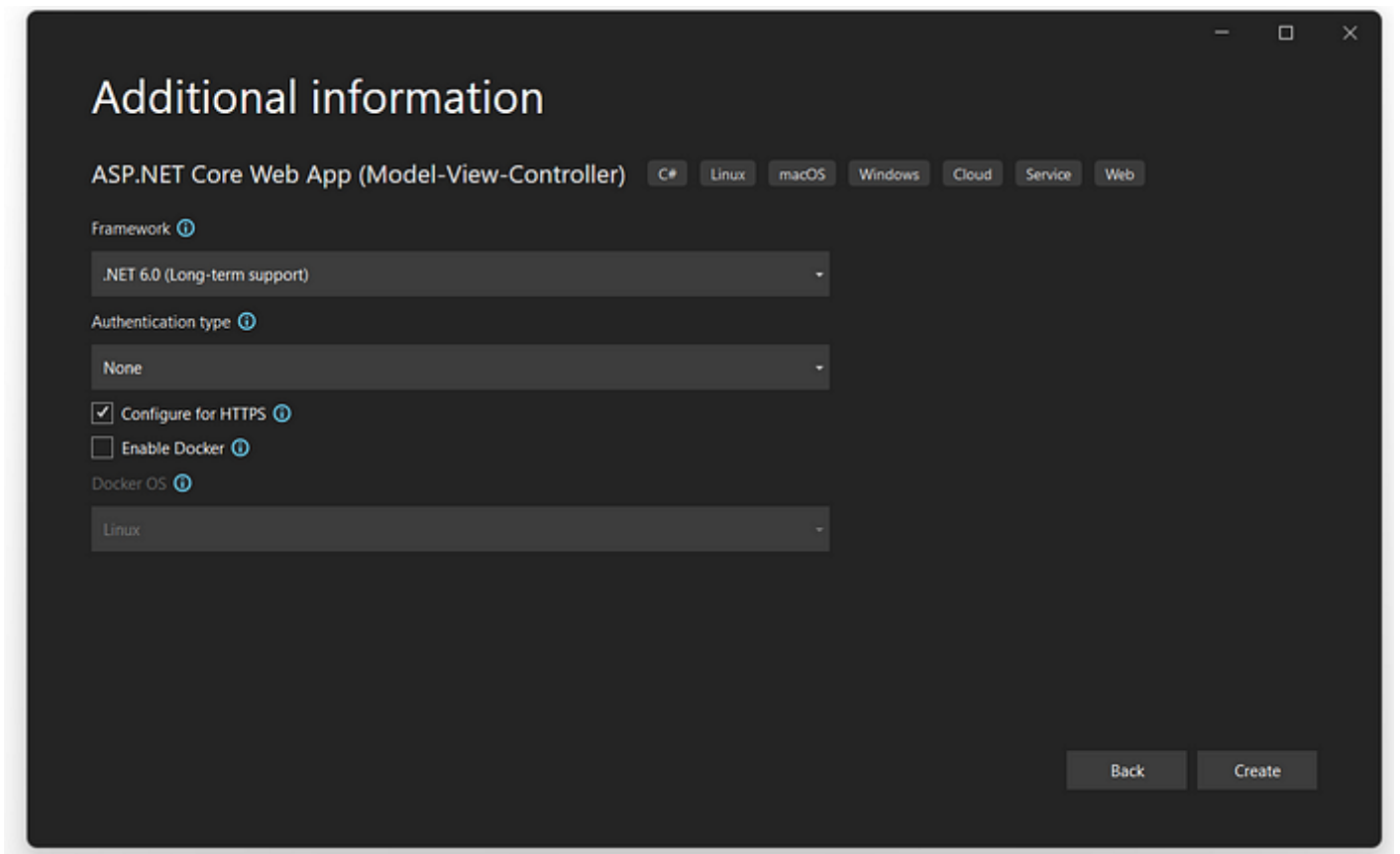
Solution name ⓘ

WebApplication1

☐ Place solution and project in the same directory

Back Next

Now, choose framework .NET 6.0. Then click the create button.



Additional information

ASP.NET Core Web App (Model-View-Controller) C# Linux macOS Windows Cloud Service Web

Framework ⓘ

.NET 6.0 (Long-term support)

Authentication type ⓘ

None

☒ Configure for HTTPS ⓘ

☐ Enable Docker ⓘ

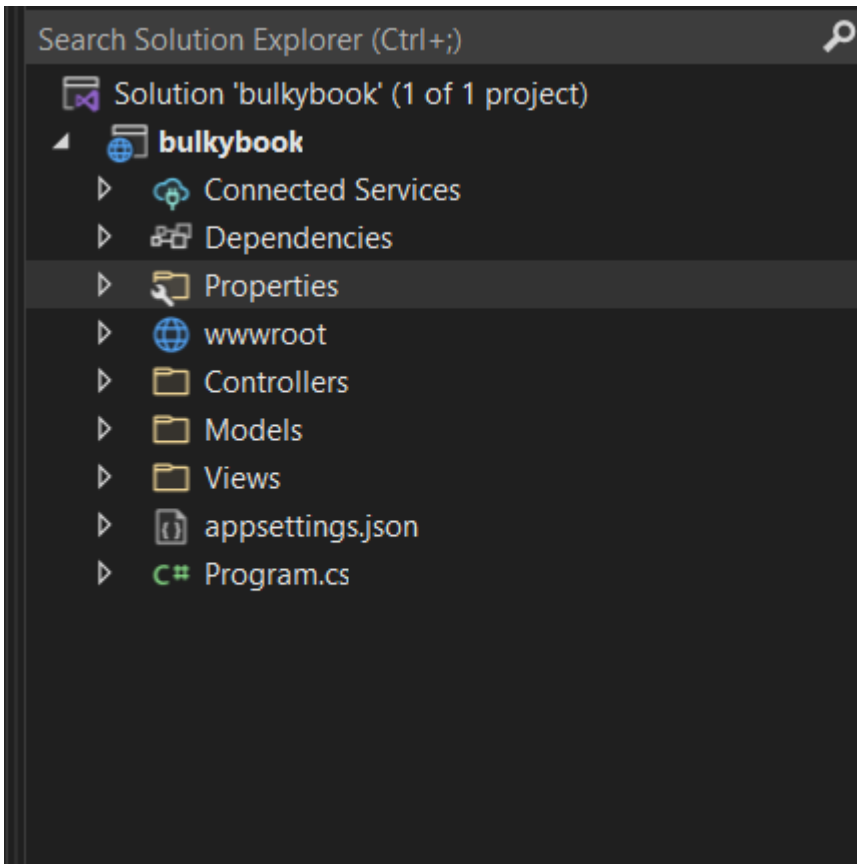
Docker OS ⓘ

Linux

Back Create

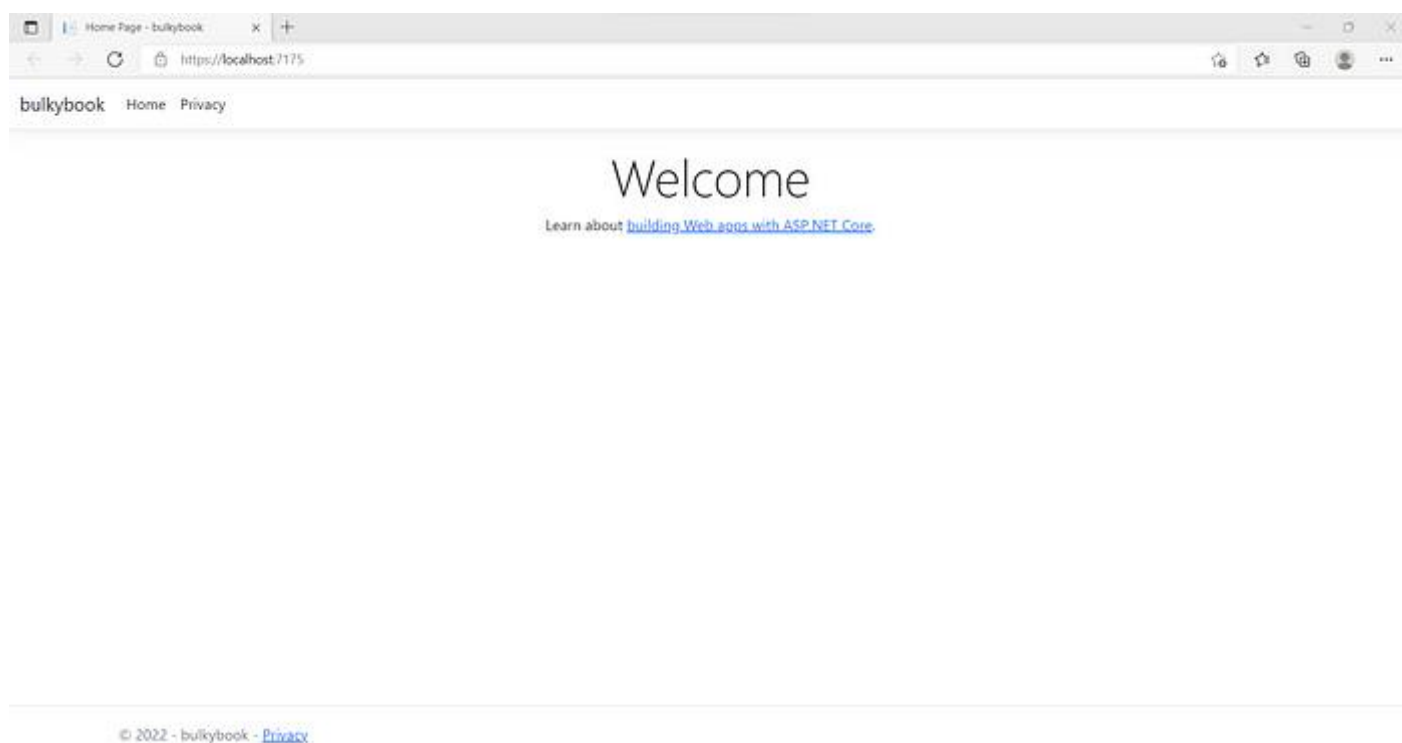
Asp.Net Core MVC 6.0 application created and project structure is shown below,





Now, build and run the application.

## OUTPUT





## Installing Flutter and Configuring Android Studio for Flutter Development

**Important: Path to install flutter: C:\src\flutter**

Step-1: Download flutter from its official website.

Downloading the Flutter SDK also downloads the compatible version of Dart.

Filename: flutter\_windows\_3.3.10-stable.zip (As on 3<sup>rd</sup> January, 2023)

Step-2: Go to C: drive and create a folder named src.

Step-3: Right-click on the file mentioned in step-1, select Open with, click WinRAR.

Extract the file in C:\src folder.

The C:\src\flutter folder must list few of the following folders/files:

.git  
.github  
.idea  
.pub-cache  
bin  
dev  
examples  
packages  
.ci.yaml

etc...

Total: 25 items

Step-4 Type env in search box and click on Edit the System Environment variables.

Step-5 In the System Properties dialog box, select Advanced tab and Click on Environment Variables button shown at the bottom.

Step-6 In the Environment Variables dialog box, look at User variables for <<Username>>, double-click on Path.

Step-7 In the Edit Environment Variables dialog box, click on New, type in C:\src\flutter\bin and Click Ok.

Step-8 Go to command prompt and run then following command:

```
flutter --version
```

It should give following output:

Flutter 3.3.9 • channel stable • <https://github.com/flutter/flutter.git>

Framework • revision b8f7f1f986 (11 days ago) • 2022-11-23 06:43:51 +0900

Engine • revision 8f2221fbef

Tools • Dart 2.18.5 • DevTools 2.15.0



## Tool Based Manual || MCA || SEM II || CA871 : Advanced Mobile Programming

C:\Users\CHARUSAT>flutter doctor

Doctor summary (to see all details, run flutter doctor -v):

[V] Flutter (Channel stable, 3.3.9, on Microsoft Windows [Version 10.0.22621.674], locale en-US)

[!] Android toolchain - develop for Android devices (Android SDK version 33.0.1)

X cmdline-tools component is missing

Run `path/to/sdkmanager --install "cmdline-tools;latest"`

See <https://developer.android.com/studio/command-line> for more details.

X Android license status unknown.

Run `flutter doctor --android-licenses` to accept the SDK licenses.

See <https://flutter.dev/docs/get-started/install/windows#android-setup> for more details.

[V] Chrome - develop for the web

[X] Visual Studio - develop for Windows

X Visual Studio not installed; this is necessary for Windows development.

Download at <https://visualstudio.microsoft.com/downloads/>.

Please install the "Desktop development with C++" workload, including all of its default components

[V] Android Studio (version 2021.3)

[V] Connected device (3 available)

[V] HTTP Host Availability

! Doctor found issues in 2 categories.

## Adding Flutter Plug-in to Android Studio

Step-1 Open Android Studio> File menu> Settings > Plugins (below Editor).

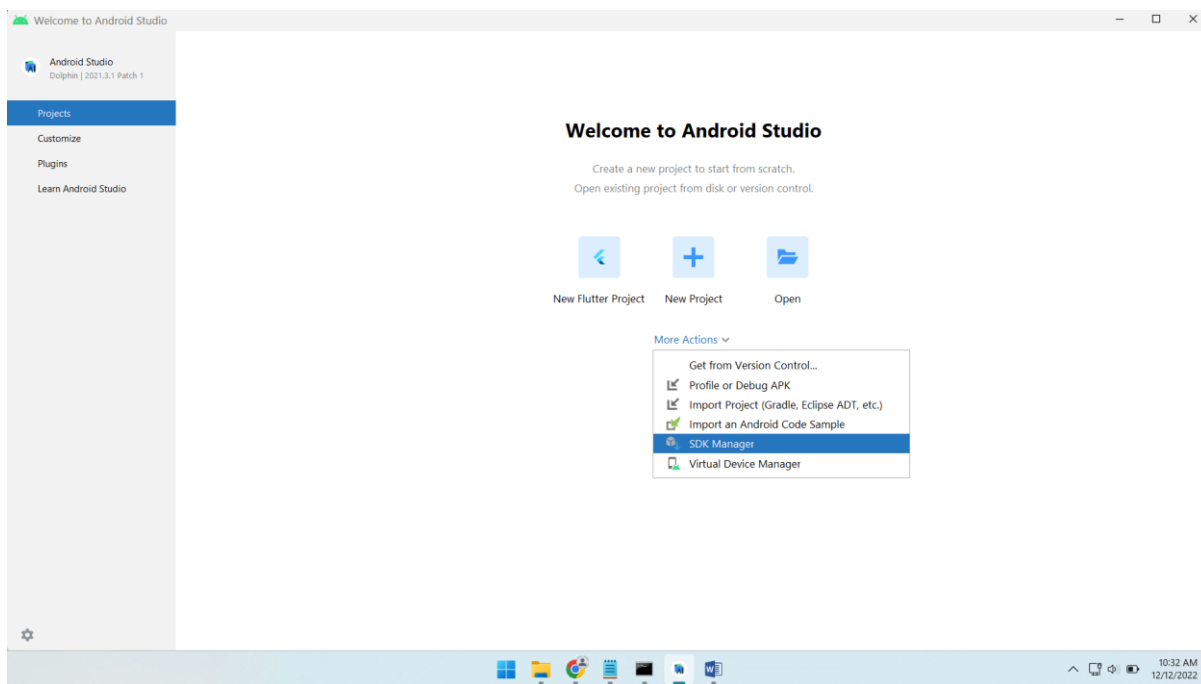
Step-2 Install: Flutter and Dart Plugins.

## Configuring Android Studio for Flutter after installation

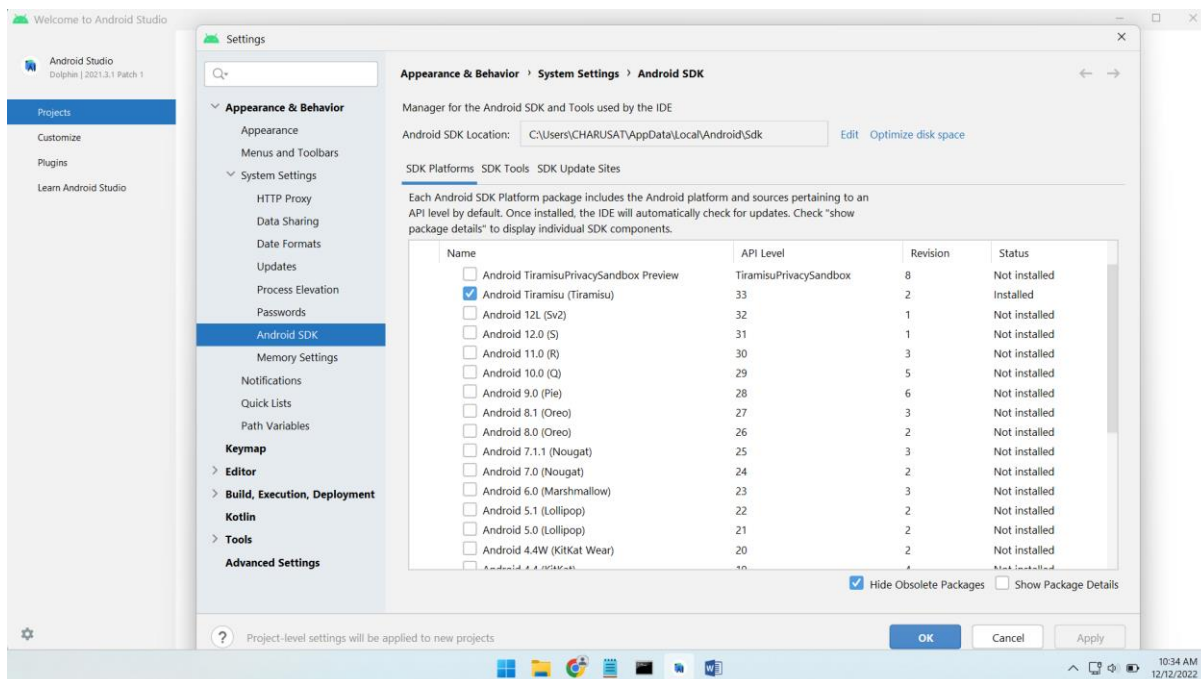
**How to resolve "cmdline-tools component is missing" issue?**

Step-1 Open Android Studio, click on More Actions drop-down and select SDK Manager.

## Tool Based Manual || MCA || SEM II || CA871 : Advanced Mobile Programming

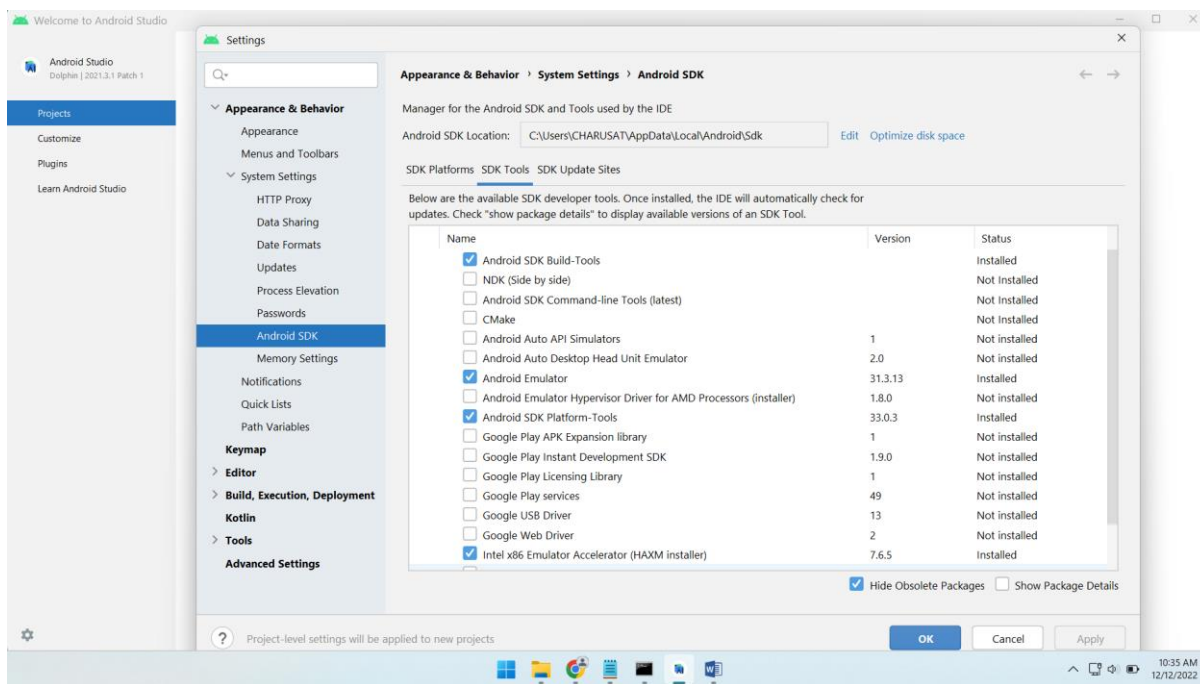


Step-2 In the Settings window, select Android SDK on LHS.

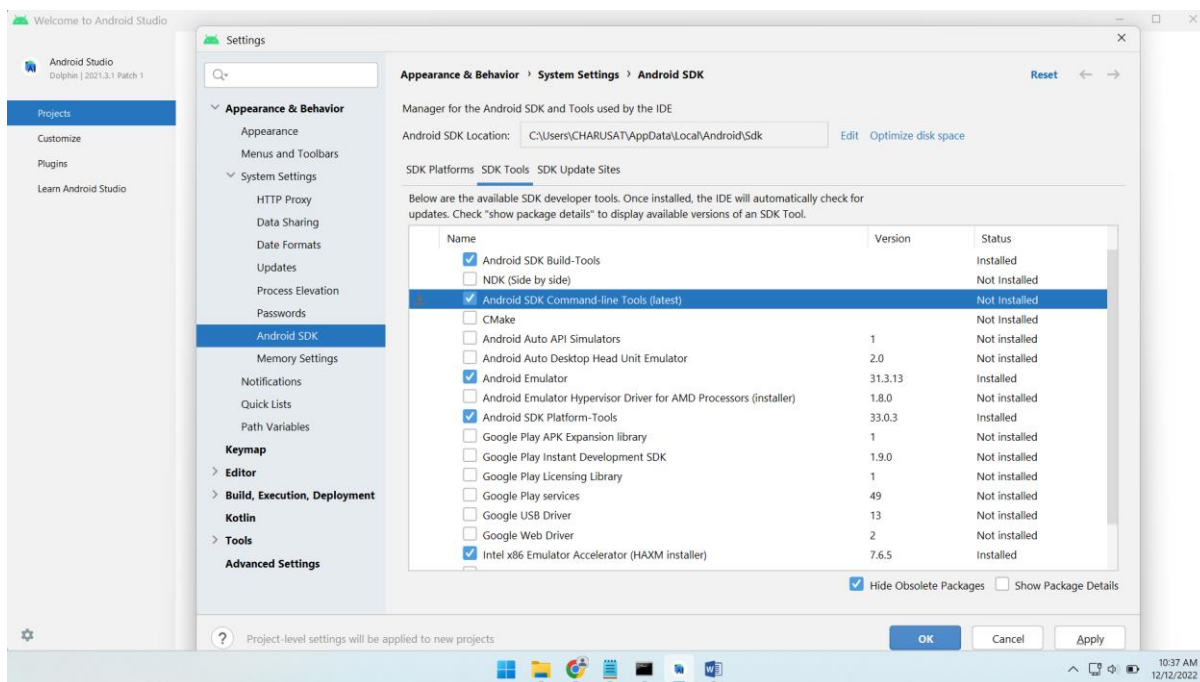


Next, click on SDK Tools tab on RHS.

## Tool Based Manual || MCA || SEM II || CA871 : Advanced Mobile Programming



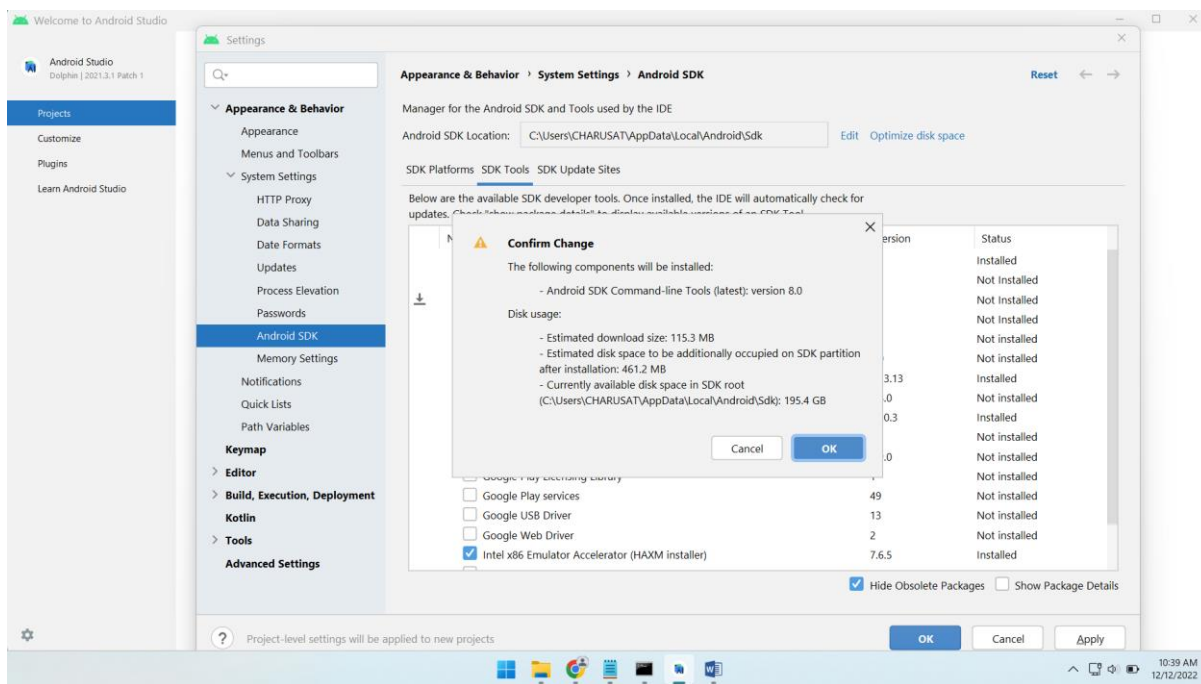
Step-3 Select Android SDK Command-line Tools(latest). Make sure you click to put the tick mark before its name. Click on Apply button.



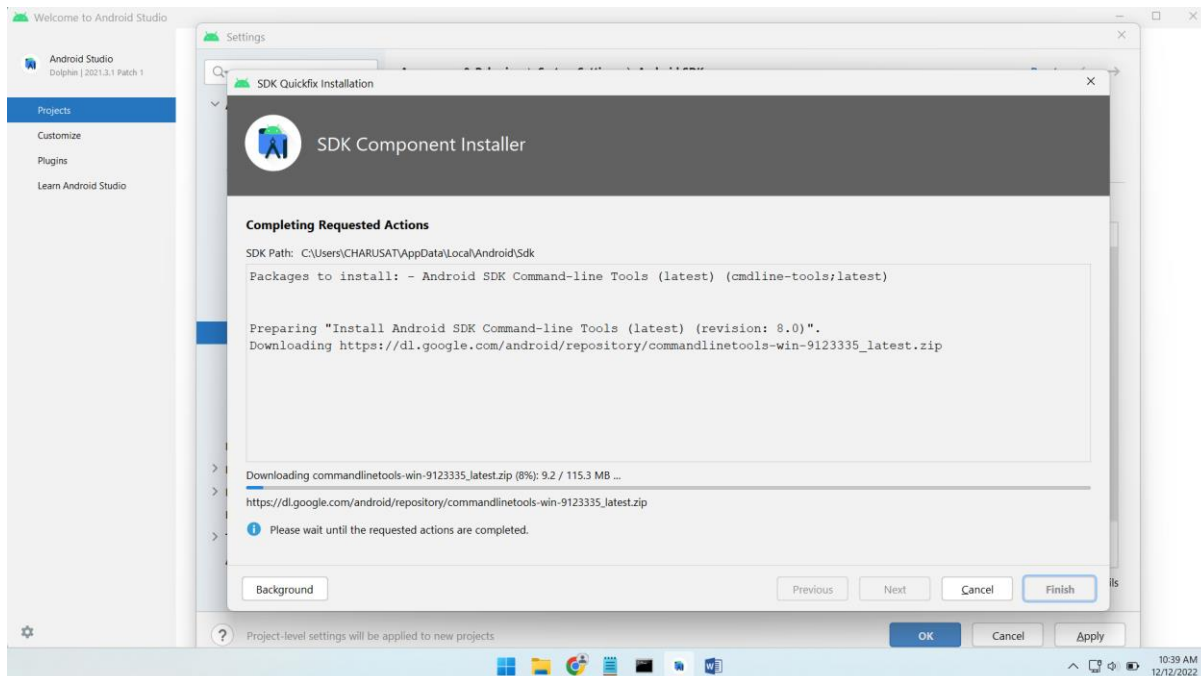
Click OK to confirm.



## Tool Based Manual || MCA || SEM II || CA871 : Advanced Mobile Programming



The installation will begin.



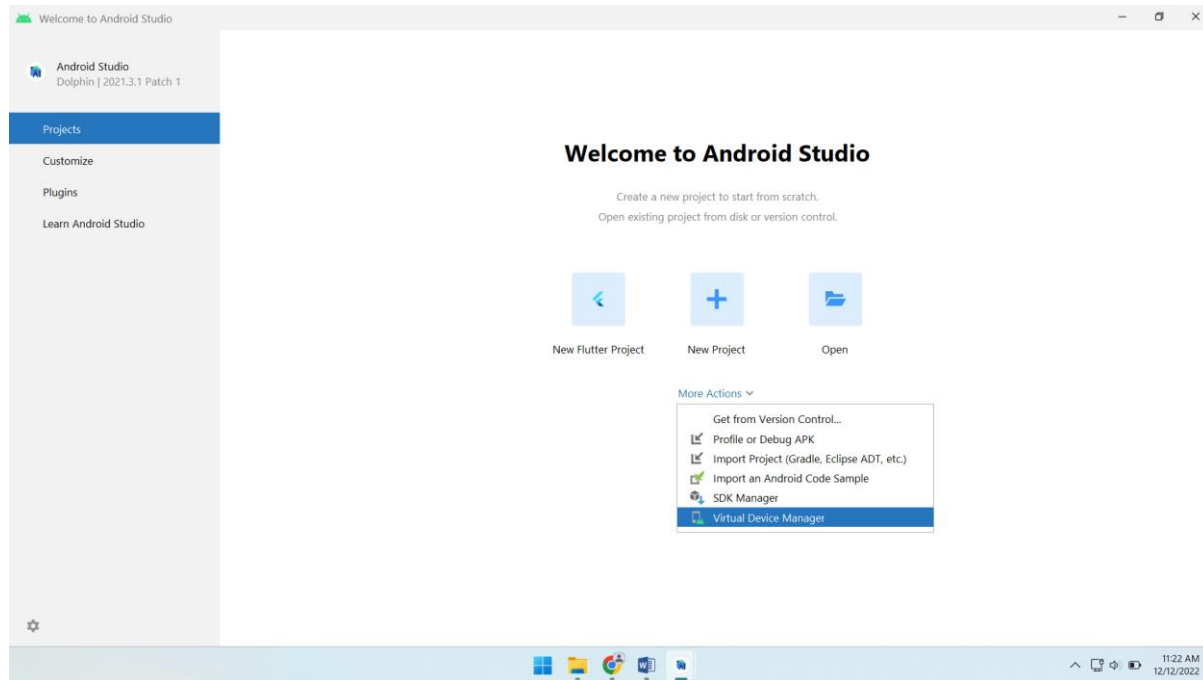
Click Finish button upon completion.

Set JAVA\_HOME variable to: C:\Program Files\Java\jdk1.8.0\_171 and not to C:\Program Files\Java\jdk1.8.0\_171\bin



## Set up the Android emulator

Step-1 Open Android Studio, click on More Actions drop-down and select Virtual Device Manager.



To change font-size in the code editor, go to File menu > Settings > Editor (on LHS) > Font > Size.

## ➤ IDE

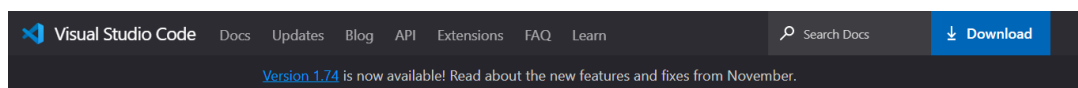
- IDE stands for **Integrated development environment**.
- It is nothing more than an enhanced version of a text editor that helps you write more efficient and nicer code.
- It helps to differentiate different parts of your codes with different colors and notifying you if you are missing some semicolon or bracket at some place by highlighting that area.
- A lot of IDEs are available such as **DEV++ Code Blocks, VS Code**.

## ➤ Visual Studio Code

- Getting up and running with Visual Studio Code is swift and straightforward.
- VS Code is a free text editor created by Microsoft.
- It is a good editor for JavaScript, HTML and CSS.
- It is a small download so you can install it quickly.
- Additionally, it runs on the Windows, macOS, and Linux operating systems.
- It also supports a large number of extensions that allow the VS Code editor to be adapted to many purposes.

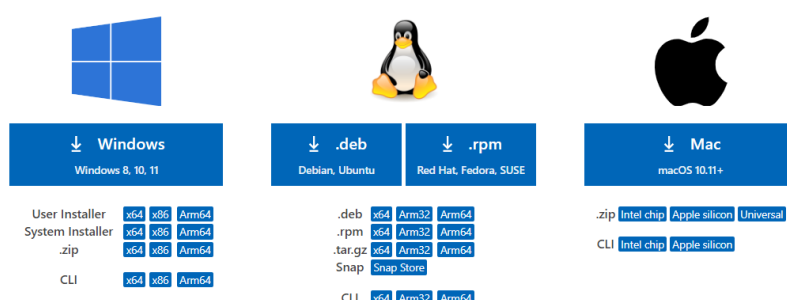
## ➤ Visual Studio Code Installation

- Visit <https://code.visualstudio.com/download>



### Download Visual Studio Code

Free and built on open source. Integrated Git, debugging and extensions.



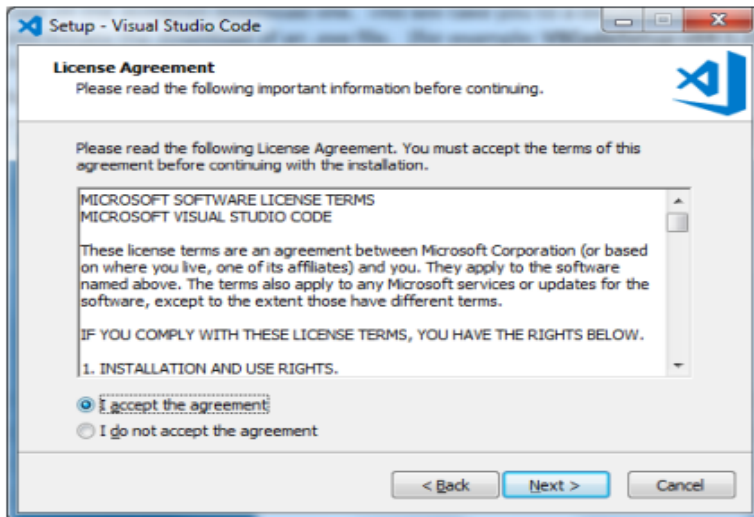
## Tool Based Manual || BCA || SEM II || CA128 : Advance Web Designing

- Click on the Windows download link. This will take you to a documentation page and also initiate the download of an .exe file.
- Run the exe. Step through the installation wizard as indicated below.

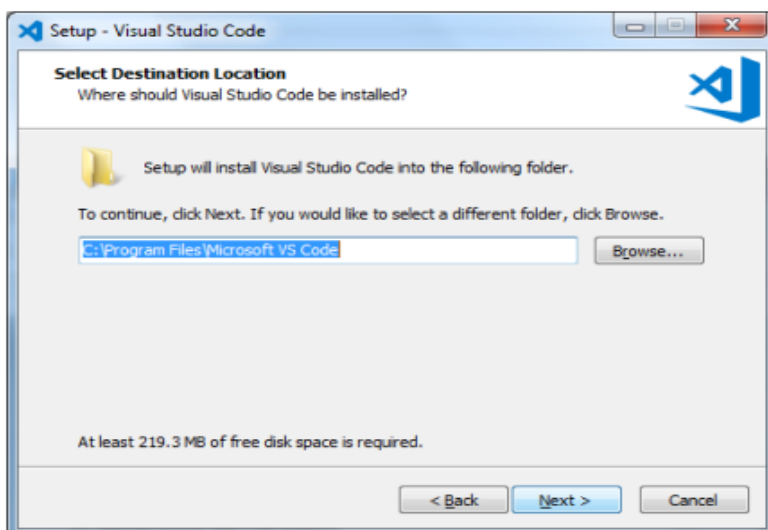


Click Next

## Tool Based Manual || BCA || SEM II || CA128 : Advance Web Designing

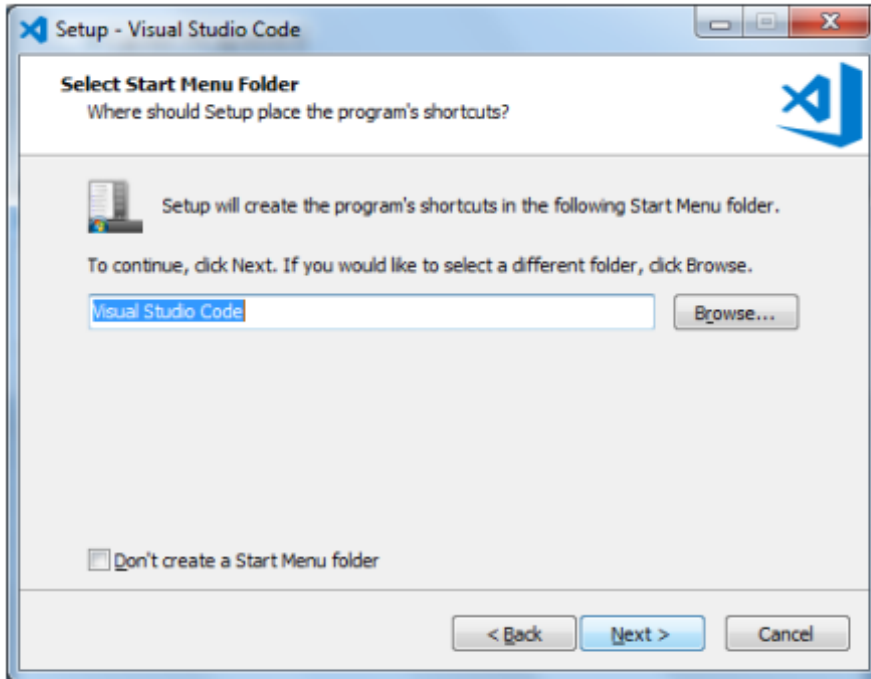


Accept the license agreement.



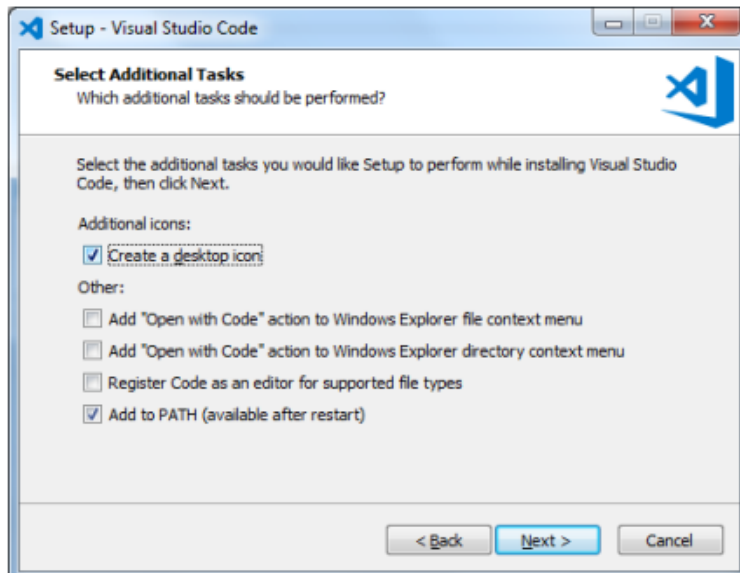
Use the default installation folder. Click Next.



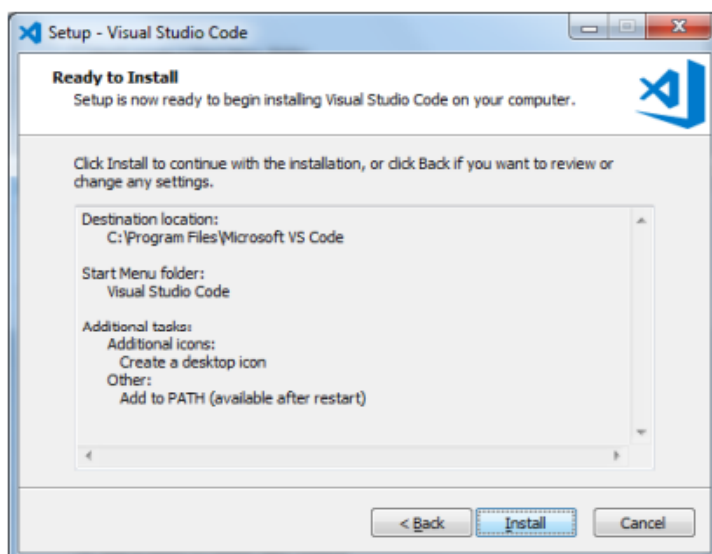


Click Next

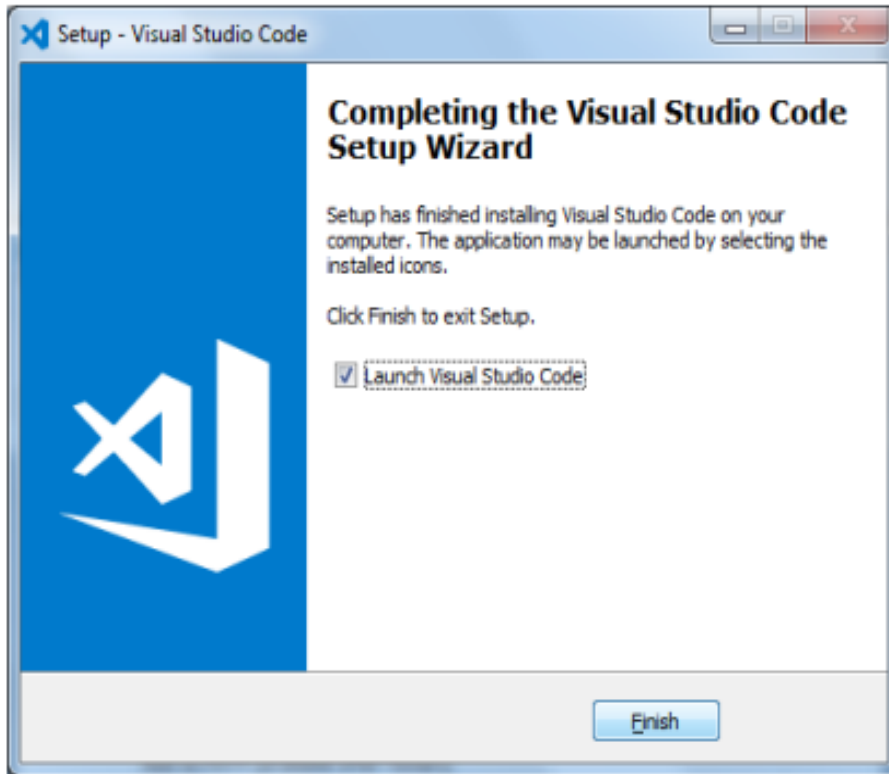
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Creating a desktop shortcut is recommended. You will use VS Code a lot in this course, and you want it to be easy to find. Check the boxes as shown. Click Next

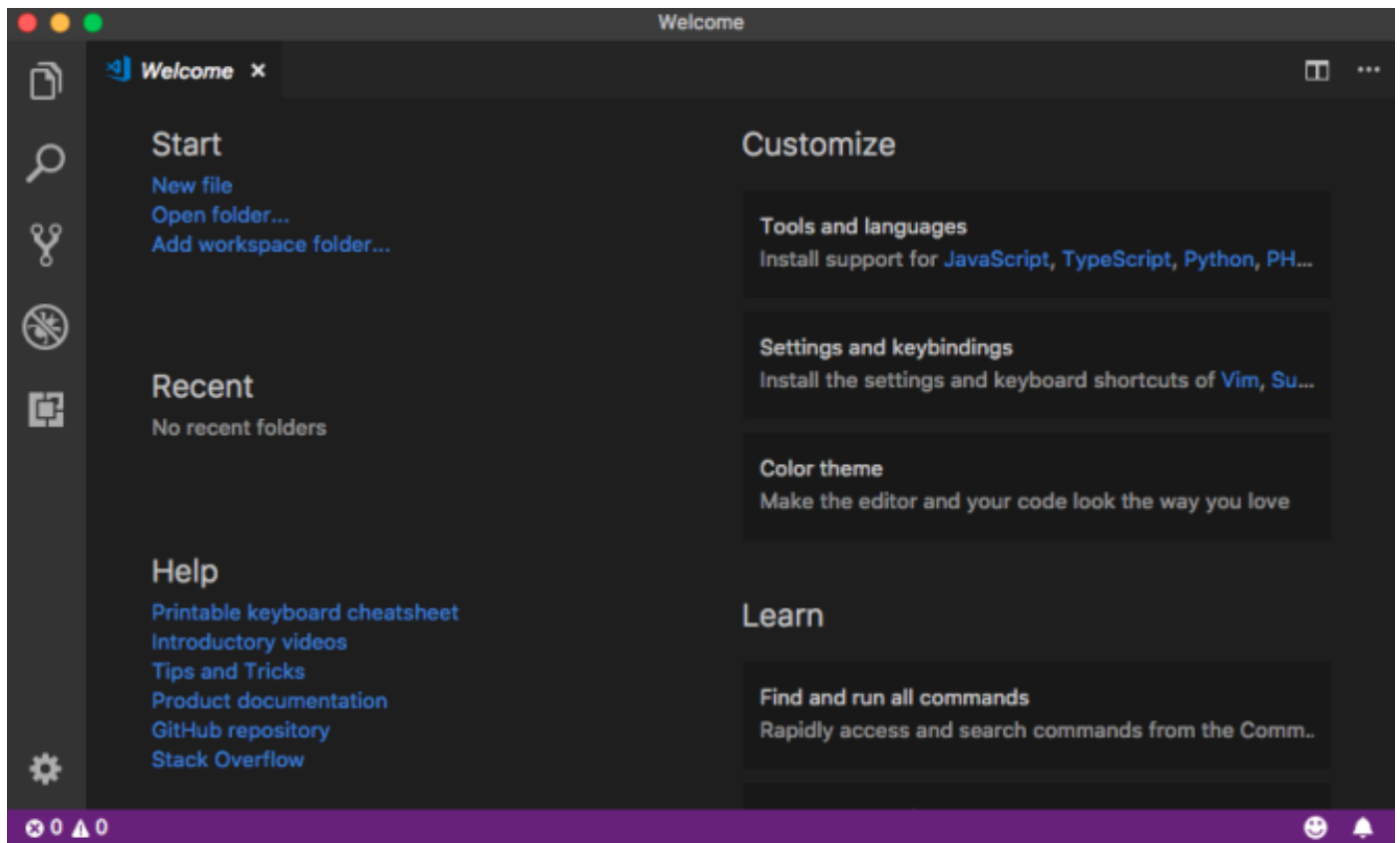


Click Install.



Click Finish.

- VS Code will start, and you will see a screen similar to the following:



- On the left side, you should see a couple of icons. One of them (under the no bugs icon) is the extensions button:

### ➤ Extensions:

Keyboard Shortcut: `Ctrl+Shift+X`

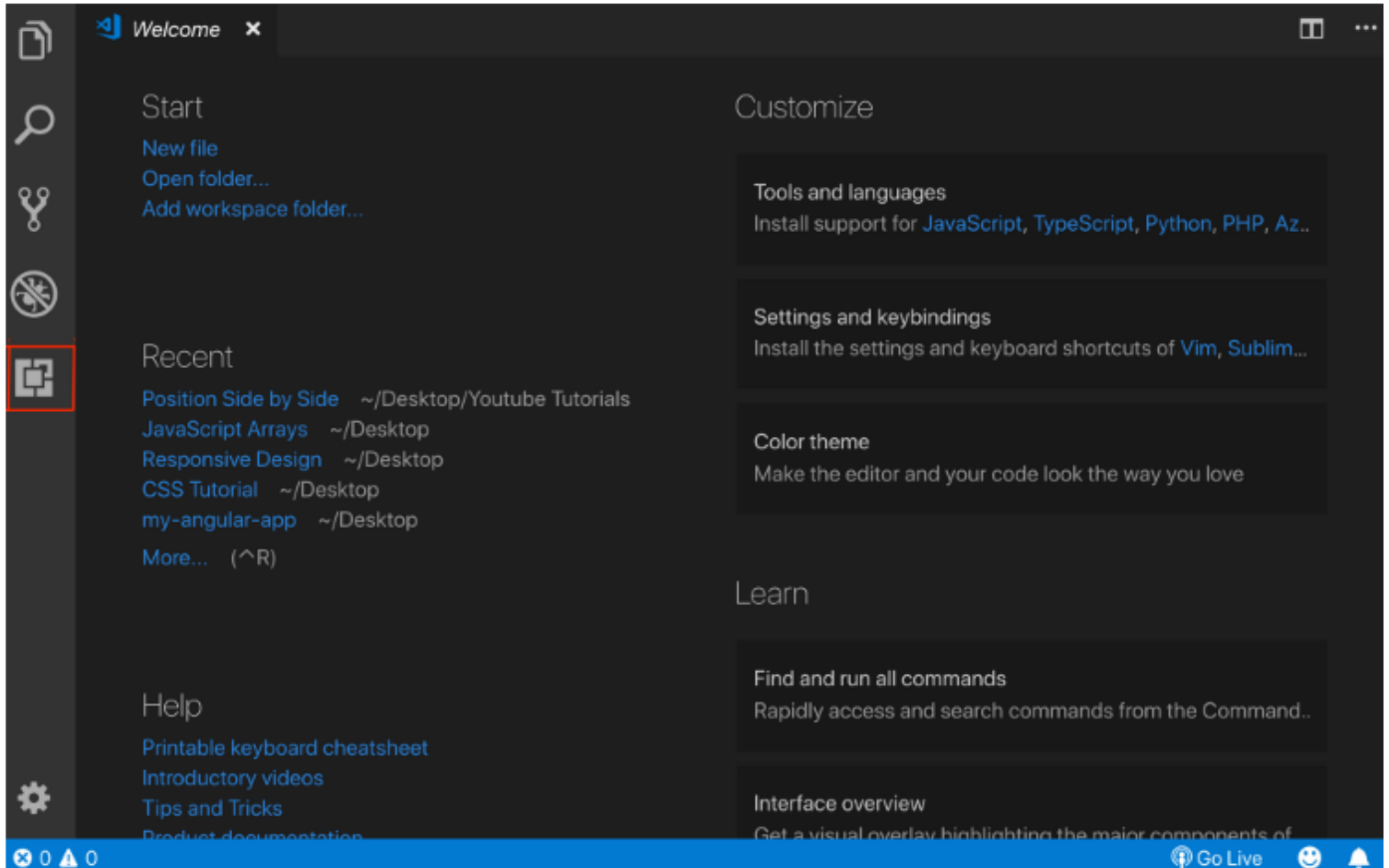
#### Find extensions:

1. In the VS Code Marketplace.
2. Search inside VS Code in the **Extensions** view.
3. View extension recommendations
4. Community curated extension lists, such as awesome-vscode

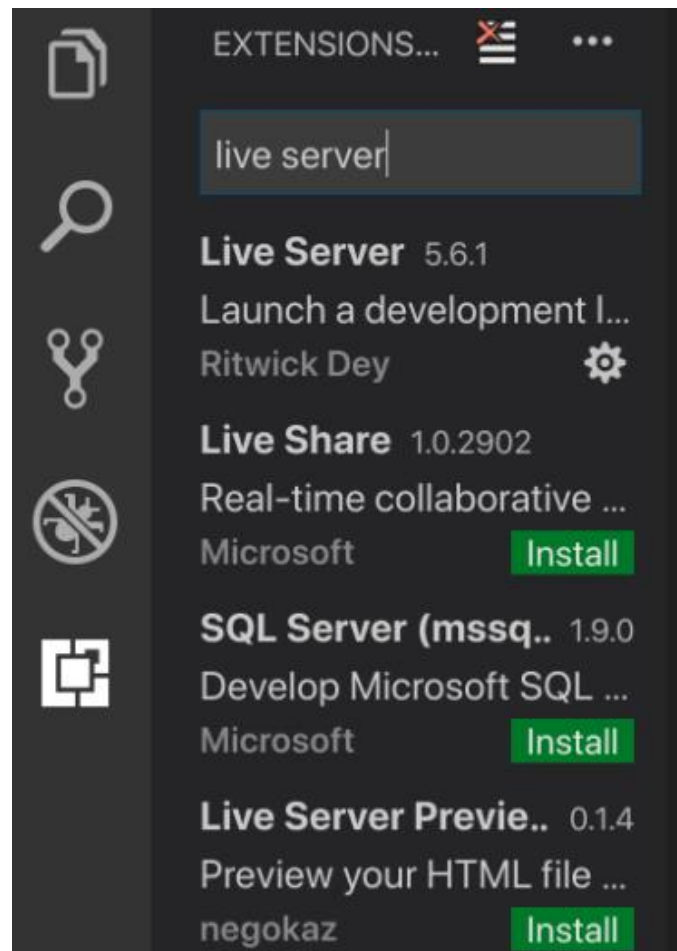
Enable additional languages, themes, debuggers, commands, and more. VS Code's growing community shares their secret sauce to improve workflow.

#### Installing Extensions:

In the **Extensions** view, you can search via the search bar or click the **More Actions (...)** button to filter and sort by install count.



- Once you click on it a search bar will appear. Just type in **“live server”**.

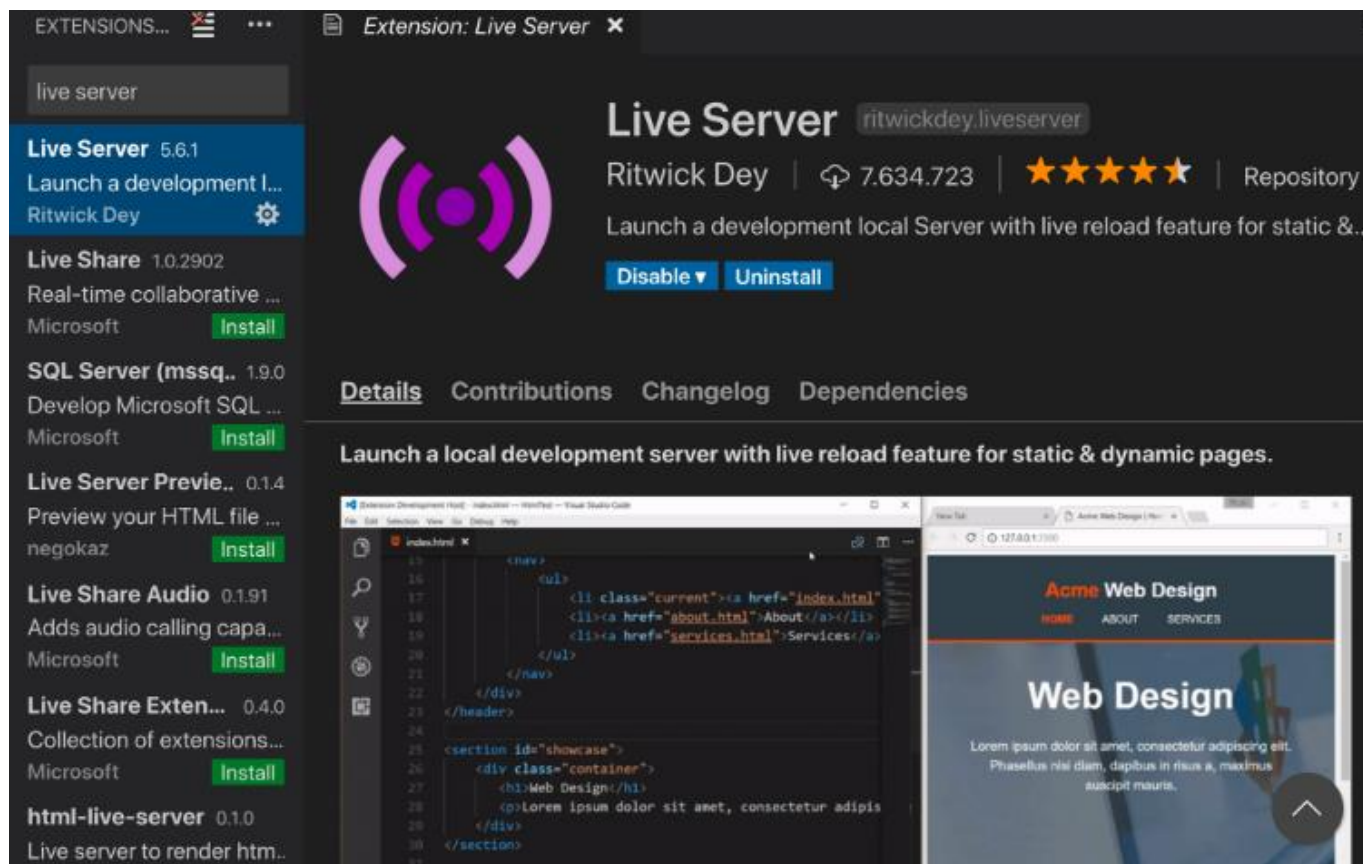


### ➤ Live Server installation

- You will see many options, so you can choose whichever one works for your system.



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### ➤ Why Live Server

- Normally, when you make a change in your code or write something new, you need to refresh the page manually to see the changes.
- In other words, if you make 100 changes in your code each day, you need to refresh the browser 100 times.
- The live-server extension, however, automates this for you. After installing it, an automated localhost will be able to run in your browser, which you can start with a single button.
- Once you make changes in your code or write something new, after saving it, the browser will auto-refresh itself. Then you will be able to see the changes quickly and automatically.



**Prepared by : Dr. Kamini Solanki**

**Id: 1061**

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## Preview

- Android is an operating system and programming platform developed by Google for mobile phones and other mobile devices, such as tablets.
- It can run on many different devices from many different manufacturers. Android includes a software development kit (SDK) that helps you write original code and assemble software modules to create apps for Android users.
- The first beta version of the Android Software Development Kit (SDK) was released by Google in 2007 where as the first commercial version, Android 1.0, was released in September 2008.
- Now the Kotlin is official language for development instead Java.



## Creating an Android Application

**Tools required:** Android Studio Version Dolphin

**Table to Download links for tools:**

|                                                                   |                                                                                         |
|-------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| <a href="https://developer.android.com/studio">Android Studio</a> | <a href="https://developer.android.com/studio">https://developer.android.com/studio</a> |
|-------------------------------------------------------------------|-----------------------------------------------------------------------------------------|

### Creating our first Android Application:

1. Open Android Studio.
2. In the Welcome to Android Studio dialog, click Start a new Android Studio project.
3. Select Empty Activity
4. Give your application a name such as My First App.
5. Make sure the Language is set to Kotlin.
6. Leave the defaults for the other fields.
7. Click Finish.